

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\LP Flash Column Overhead Reflux Drum (153-D)Rev 07 percent.pvdb

Date of Analysis : Sep 16,2026 1:54pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 4506.000 mm. ID	
Vessel Design Length, Tangent to Tangent	4500.00 mm.
Distance of Bottom Tangent above Grade	5300.00 mm.
Distance of Base above Grade	300.00 mm.
Specified Datum Line Distance	5000.00 mm.
Internal Design Temperature	110 °C
Internal Design Pressure	3.500 bars
External Design Temperature	110 °C
External Design Pressure	0.517 bars
Maximum Allowable Working Pressure	3.5 bars
External Max. Allowable Working Pressure	0.61 bars
Shop Test Pressure	6.323 bars
Field Test Pressure	6.650 bars
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-48.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-36	...	...	K02600	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-182 F316L	...	> 130	S31603	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F316L	...	> 130	S31603	
Rings	SA-516 70	...	...	K02700	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-307 A	...	6 <= t <= 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt	...	...	...	1.0000	N/A	No
Skirt	...	...	...	1.0000	N/A	No
Skirt	...	...	...	1.0000	N/A	No
BOTTOM HEAD	3.679	0.52	4.510	1.0000	No	No
SHELL-1	3.565	0.52	5.640	1.0000	N/A	No
SHELL-2	3.507	0.52	5.700	1.0000	N/A	No
TOP HEAD	3.507	0.52	4.680	1.0000	No	No

Liquid Level: 1776.50 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.987

Stiffener Ring Specifications:



Elevation mm.	Selected Type	Ring Description
550.00	Bar 250.0 x 12.0	Ring R1 Fr50
3750.00	Bar 250.0 x 12.0	Ring R1 Fr60

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	-3000.0	2000.0	12.0	12.0	...	...	0.70	0.70
Skirt	-1000.0	2000.0	12.0	12.0	...	...	0.70	0.70
Skirt	0.0	1000.0	12.0	12.0	...	...	0.70	0.50
Ellipse	50.0	50.0	12.0	10.0	8.1	...	0.85	0.85
Cylinder	3050.0	3000.0	12.0	12.0	7.9	...	0.85	0.85
Cylinder	4450.0	1400.0	12.0	12.0	7.8	...	0.85	0.85
Ellipse	4500.0	50.0	12.0	10.0	7.7	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	10	9.28625	0.51708	0.61
50	Ring	12	6.69686	0.51708	2.6
Ring	60	12	10.5524	0.51708	0.73
60	Ring	12	10.5524	0.51708	0.73
Ring	70	12	7.14708	0.51708	2.24
70	80	10	9.28625	0.51708	0.61

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	5000	No Calc	No Calc	No Calc
20	30	5000	No Calc	No Calc	No Calc
30	40	5000	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	Ring	925.5	4497.64	No Calc	No Calc
Ring	60	3200	4497.86	681.956	3985.79
60	Ring	3200	4497.86	No Calc	No Calc
Ring	70	1125.5	4498.06	716.2	3985.79
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3-4"	WRC-107/537	0.601	n/a	Passed
N1-42"	WRC-107/537	2.086	n/a	Failed
N4-2"	WRC-297	0.556	Shell	Passed
N2-42"	WRC-107/537	1.928	n/a	Failed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kg	Earthquake Shear Kg	Wind Bending Kg-m.	Earthquake Bending Kg-m.
------	----	-------------------------	--------------------------	---------------------	--------------------	--------------------------

10	20	1000	3538.72	11176.5	22840	91180.7
20	30	3000	3088.67	11090.3	16212.5	68913.5
30	40	4500	2638.62	10936.5	10485.1	46886.3
40	50	5025	2413.59	10821.1	7958.95	36007.3
50	60	6550	2145.76	8893.77	7844.97	35514.4
60	70	8750	1470.08	6696.62	2421.1	12128.4
70	80	9475	1154.77	6172.48	583.67	3119.84

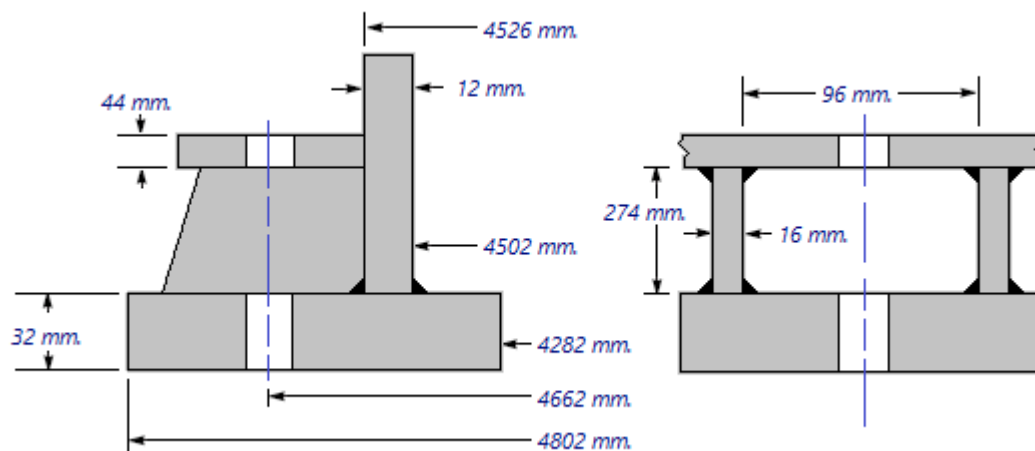
Abs Max of the all of the Stress Ratio's : 0.3758

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	32.0000	mm.
Inside Diameter of Basing	4282.0000	mm.
Outside Diameter of Basing	4802.0000	mm.
Nominal Diameter of Bolts	48.0000	mm.
Diameter of Bolt Circle	4662.0000	mm.
Number of Bolts	20	

Thickness of Gusset Plates	16.0000	mm.
Average Width of Gusset Plates	130.0000	mm.
Height of Gussets	274.0000	mm.
Distance between Gussets	96.0000	mm.
Thickness of Top Plate or Ring	44.0000	mm.
Radial Width of the Top Plate	140.0000	mm.
Number of Gussets	40	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	11867.	Kgf
Total Earthquake Shear on Support	19505.	Kgf
Wind Moment on Support	135616.	Kg-m.
Earthquake Moment on Support	203957.	Kg-m.
Maximum Initial Bolt Load	4033.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	27853.5	kg.
Shop Test - Fabricated + water (full)	137874.7	kg.
Shipping - Fab. + rem. intls.+ shipping app.	27853.5	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	65760.5	kg.



Empty	- Fab. + intls. + details + wghts.	65760.5 kg.
Operating	- Empty + operating liquid (no ca)	87727.4 kg.
Field Test	- Empty weight + test medium (full)	161095.2 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3-4"	3.5	OK	...	OK	6.26	OK	Passed
N1-42"	3.5	OK	...	OK	6.74	OK	Passed
N4-2"	3.5	OK	...	OK	5.80	OK	Passed
K1-2"	3.5	OK	...	OK	5.80	OK	Passed
K2-2"	3.5	OK	...	OK	5.80	OK	Passed
K4-2"	3.5	OK	...	OK	5.80	OK	Passed
K5-2"	3.5	OK	...	OK	5.80	OK	Passed
K3-2:	3.5	OK	...	OK	5.80	OK	Passed
K6-2"	3.5	OK	...	OK	5.80	OK	Passed
M1-24"	3.5	OK	...	...		OK	Passed
N2-42"	3.5	OK	...	OK	6.73	OK	Passed
K7-2"	3.5	OK	...	OK	5.80	OK	Passed
K8-2"	3.5	OK	...	OK	5.80	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 3.5 Nozzle : K8-2"
 Minimum MAWP Shells/Flanges : 3.5 Element : BOTTOM HEAD
 Minimum MAPnc Shells/Flanges : 5.2 Element : TOP HEAD

Computed Vessel M.A.W.P. : 3.5 bars

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
40	N3-4"	0.000	0.000	188.099
50	N1-42"	2050.000	90.000	2137.600
50	N4-2"	900.000	225.000	106.328
50	K1-2"	1850.000	270.000	106.328
50	K2-2"	300.000	270.000	106.328
50	K4-2"	300.000	45.000	106.328
50	K5-2"	900.000	135.000	106.328
50	K3-2:	900.000	270.000	106.328
50	K6-2"	300.000	225.000	106.328
50	M1-24"	2050.000	0.000	1223.200
70	N2-42"	0.000	0.000	2137.600
70	K7-2"	0.000	270.000	105.600
70	K8-2"	0.000	90.000	105.600

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N4-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	172.39	600
K1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	172.39	300
K2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	172.39	300
K4-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	172.39	300
K5-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	172.39	300
K3-2:	2.000 in	Actual	LWN	3.320	16.764	...	...	172.39	300
K6-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	172.39	300
K7-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	777.78	300
K8-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	777.78	300
N3-4"	4.000 in	120	WNF	4.500	11.125	220.00	10.00	260.40	300
M1-24"	24.000 in	Actual	WNF	24.945	12.000	930.00	12.00	384.38	150
N1-42"	42.000 in	Actual	WNF	42.945	12.000	1450.80	12.00	4362.00	150
N2-42"	42.000 in	Actual	WNF	42.787	10.000	1950.00	10.00	421.50	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N4-2"	SA-182 F316L	14.000	10.000	...	...	...
K1-2"	SA-182 F316L	14.000	10.000	...	...	...
K2-2"	SA-182 F316L	14.000	10.000	...	...	...
K4-2"	SA-182 F316L	14.000	10.000	...	...	...
K5-2"	SA-182 F316L	14.000	10.000	...	...	...
K3-2:	SA-182 F316L	14.000	10.000	...	...	...
K6-2"	SA-182 F316L	14.000	10.000	...	...	...
K7-2"	SA-182 F316L	10.000	10.000	...	...	...
K8-2"	SA-182 F316L	10.000	10.000	...	...	...
N3-4"	SA-106 B	10.000	10.000	10.000	10.000	...
M1-24"	SA-516 70	12.000	10.000	12.000	12.000	...
N1-42"	SA-516 70	12.000	10.000	12.000	12.000	...
N2-42"	SA-516 70	10.000	10.000	10.000	10.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
N4-2"	900.000	225.0	160.00	0.00	SHELL-1
K1-2"	1850.000	270.0	160.00	0.00	SHELL-1
K2-2"	300.000	270.0	160.00	0.00	SHELL-1
K4-2"	300.000	45.0	160.00	0.00	SHELL-1
K5-2"	900.000	135.0	160.00	0.00	SHELL-1
K3-2:	900.000	270.0	160.00	0.00	SHELL-1
K6-2"	300.000	225.0	160.00	0.00	SHELL-1
K7-2"	...	270.0	750.00	0.00	TOP HEAD
K8-2"	...	90.0	750.00	0.00	TOP HEAD
N3-4"	...	0.0	250.00	0.00	BOTTOM HEAD
M1-24"	2050.000	0.0	350.00	0.00	SHELL-1

PV Elite 28 U1 Licensee: SL User
FileName : LP Flash Column Overhead Reflux Drum (153-D)_Rev Page 8 of 167
Nozzle Schedule: Step: 41 1:55pm Sep 16,2026



HEXAGON

N1-42"	2050.000	90.0	350.00	4000.00	SHELL-1
N2-42"	...	0.0	375.00	0.00	TOP HEAD

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm. THK X 4506.0mm. ID X 50.0mm.	SA-516 70
1	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 5000.0mm	...
2	CLADDING: 1176.500mm X 3.000mm THK	...
1	CLADDING: 3000.000mm X 3.000mm THK	...
2	BAR RING STIFFENER: 4530.0mm ID X 5030.0mm OD X 12.0mm	SA-516 70
1	CLADDING: 1400.000mm X 3.000mm THK	...
1	PLATFORM: 1000mm X 1000mm WIDE	
1	LADDER: 5850.0mm	...
1	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 4802.0mm. OD X 4282.0 ID X 32.0mm.	SA-516 70
20	BASERING BOLTS: 48.0mm. DIA	SA-307 A
20	BASERING BOLT NUTS: 48.0mm. DIA	...
1	BASERING TOP RING: 4806.0mm. OD X 4526.0 ID X 44.0mm.	...
40	BASERING GUSSET PLATES: 130.0mm. X 130.0mm. X 274.0mm. X 16.0mm.	...
8	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 3045.2mm. X 2000.0mm.	SA-36
2	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 1962.2mm. X 2000.0mm.	SA-36
4	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 3045.2mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 1962.2mm. X 1000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 12.0mm. THK X 5632.4mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 2891.0mm. X 3000.0mm.	SA-516 70
4	CYLINDRICAL SEGMENT 12.0mm. THK X 3048.0mm. X 1400.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 1964.0mm. X 1400.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
BOTTOM HEAD	[10]	D	-48	-48		0.635	10.000	0.85	No
BOTTOM HEAD	[7]	D	-48	-48		0.521	12.000	0.85	No
SHELL-1	[8]	D	-48	-48		0.521	12.000	0.85	No
SHELL-2	[8]	D	-48	-48		0.521	12.000	0.85	No
TOP HEAD	[10]	D	-48	-48		0.635	10.000	0.85	No
TOP HEAD	[7]	D	-48	-48		0.521	12.000	0.85	No
Ring R1 Fr50	[23]	B	-23	-48		0.521	12.000	0.85	No
Ring R1 Fr60	[23]	B	-23	-48		0.521	12.000	0.85	No
N3-4"	[1]	B	-29	-48		0.601	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-42"	[1]	D	-48	-48		0.522	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M1-24"	[1]	D	-48	-48		0.522	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N2-42"	[1]	D	-48	-48		0.573	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No
Warmest MDMT:			-18	-48					
Required Minimum Design Metal Temperature						-12.0	°C		
Warmest Computed Minimum Design Metal Temperature						-48.0	°C		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	3.4998	bars
Design Internal Temperature	110.0	°C
Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-3	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW
Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.
Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000
M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	Skirt	
Distance "FROM" to "TO"	2000	mm.
Skirt Inside Diameter	4502	mm.
Diameter of Skirt at Base	4502	mm.
Skirt Thickness	12	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Temperature Internal Pressure	110	°C
Design Temperature External Pressure	110	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-36	
Allowable Stress, Ambient	114	N./mm ²
Allowable Stress, Operating	114	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	225.4	N./mm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-2	
UNS Number	K02600	
Product Form	Plate; sheet	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element#	2/7	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt	
Distance "FROM" to "TO"	2000	mm.
Skirt Inside Diameter	4502	mm.
Diameter of Skirt at Base	4502	mm.
Skirt Thickness	12	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Temperature Internal Pressure	110	°C
Design Temperature External Pressure	110	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-36	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element#	3/7	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	Skirt	
Distance "FROM" to "TO"	1000	mm.
Skirt Inside Diameter	4502	mm.
Diameter of Skirt at Base	4502	mm.
Skirt Thickness	12	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Temperature Internal Pressure	110	°C
Design Temperature External Pressure	110	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²



Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	237.4	N./mm ²
UCS-66 Chart Curve Designation	D	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element#	4/7	
Element From Node	40	
Element To Node	50	
Element Type	Elliptical	
Description	BOTTOM HEAD	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	4506	mm.
Element Thickness	10	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.4998	bars
Design Temperature Internal Pressure	110	°C
Design External Pressure	0.5171	bars
Design Temperature External Pressure	110	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	360.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	30000	cm ²
Platform Weight	4985.1	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	85	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	5000	mm.
Unit Weight of Ladder	0.003773	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	-1126.5	mm.
Height/Length of Liquid	1176.5	mm.
Liquid Density	0.0009854	kg./cm ³

Element From Node	40	
Detail Type	Lining	
Detail ID	SS 304L CLAD	
Dist. from "FROM" Node / Offset dist	-1126.5	mm.
Height/Length of Lining	1176.5	mm.
Thickness of Lining	3	mm.
Density	0.008	kg./cm ³



Element From Node	40	
Detail Type	Nozzle	
Detail ID	N3-4"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	4	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	96.839	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Weight	
Detail ID	N3-Anti Swirl B	
Dist. from "FROM" Node / Offset dist	-1125	mm.
Miscellaneous Weight	25	Kgf
Offset from Element Centerline	0	mm.

Element#	5/7	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	SHELL-1	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	4506	mm.
Element Thickness	12	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.4998	bars
Design Temperature Internal Pressure	110	°C
Design External Pressure	0.5171	bars
Design Temperature External Pressure	110	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	600	mm.
Liquid Density	0.0009854	kg./cm ³

Element From Node	50	
Detail Type	Lining	
Detail ID	Lining	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Lining	3000	mm.
Thickness of Lining	3	mm.
Density	0.008	kg./cm ³

Element From Node	50	
Detail Type	Ring	
Detail ID	Ring R1 Fr50	
Dist. from "FROM" Node / Offset dist	500	mm.
Inside Diameter of Ring	4530	mm.
Thickness of Ring	12	mm.
Outside Diameter of Ring	5030	mm.
Material Name	SA-516 70	
Height of Section Ring	0	mm.
Using Custom Stiffener Section	No	



Element From Node	50
Detail Type	Nozzle
Detail ID	N1-42"
Dist. from "FROM" Node / Offset dist	2000 mm.
Nozzle Diameter	42 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1602.2 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	50
Detail Type	Nozzle
Detail ID	N4-2"
Dist. from "FROM" Node / Offset dist	850 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	225.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.153 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L

Element From Node	50
Detail Type	Nozzle
Detail ID	K1-2"
Dist. from "FROM" Node / Offset dist	1800 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	9.6718 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L

Element From Node	50
Detail Type	Nozzle
Detail ID	K2-2"
Dist. from "FROM" Node / Offset dist	250 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	9.6718 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L

Element From Node	50
Detail Type	Nozzle
Detail ID	K4-2"
Dist. from "FROM" Node / Offset dist	250 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	45.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	9.6718 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L

Element From Node	50
Detail Type	Nozzle
Detail ID	K5-2"
Dist. from "FROM" Node / Offset dist	850 mm.
Nozzle Diameter	2 in.



Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	135.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.6718	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K3-2:	
Dist. from "FROM" Node / Offset dist	850	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.6718	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K6-2"	
Dist. from "FROM" Node / Offset dist	250	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	225.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.6718	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	M1-24"	
Dist. from "FROM" Node / Offset dist	2000	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	412.68	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	50	
Detail Type	Weight	
Detail ID	M1 Davit	
Dist. from "FROM" Node / Offset dist	2000	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	2614	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	Clip Weight	
Dist. from "FROM" Node / Offset dist	1500	mm.
Miscellaneous Weight	1144	Kgf
Offset from Element Centerline	0	mm.

Element#	6/7	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	SHELL-2	
Distance "FROM" to "TO"	1400	mm.
Inside Diameter	4506	mm.



Element Thickness	12 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.4998 bars
Design Temperature Internal Pressure	110 °C
Design External Pressure	0.5171 bars
Design Temperature External Pressure	110 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	60
Detail Type	Lining
Detail ID	SS 304 L CLAD
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	1400 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm ³

Element From Node	60
Detail Type	Ring
Detail ID	Ring R1 Fr60
Dist. from "FROM" Node / Offset dist	700 mm.
Inside Diameter of Ring	4530 mm.
Thickness of Ring	12 mm.
Outside Diameter of Ring	5030 mm.
Material Name	SA-516 70
Height of Section Ring	0 mm.
Using Custom Stiffener Section	No

Element From Node	60
Detail Type	Weight
Detail ID	Demister
Dist. from "FROM" Node / Offset dist	700 mm.
Miscellaneous Weight	350 Kg
Offset from Element Centerline	0 mm.

Element From Node	60
Detail Type	Weight
Detail ID	Demisting Suppo
Dist. from "FROM" Node / Offset dist	700 mm.
Miscellaneous Weight	121 Kg
Offset from Element Centerline	0 mm.

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	TOP HEAD
Distance "FROM" to "TO"	50 mm.
Inside Diameter	4506 mm.
Element Thickness	10 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.4998 bars
Design Temperature Internal Pressure	110 °C
Design External Pressure	0.5171 bars
Design Temperature External Pressure	110 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No



Element From Node	70	
Detail Type	Platform	
Detail ID	SQ PLT	
Dist. from "FROM" Node / Offset dist	1500	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	100000	cm ²
Platform Weight	17031	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	10000	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	5000	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	-1850	mm.
Ladder End Elevation	4000	mm.
Unit Weight of Ladder	0.003773	Kgf/mm.
Platform Length (top head platform)	10000	mm.

Element From Node	70	
Detail Type	Lining	
Detail ID	SS 304L CLAD	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Lining	1176.5	mm.
Thickness of Lining	3	mm.
Density	0.008	kg./cm ³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	N2-42"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	42	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	452.85	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	K7-2"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	26.41	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	K8-2"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	26.41	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	70	
Detail Type	Weight	



Detail ID	PSV Weight-1	
Dist. from "FROM" Node / Offset dist	1176	mm.
Miscellaneous Weight	3500	Kgf
Offset from Element Centerline	5000	mm.
Element From Node	70	
Detail Type	Weight	
Detail ID	PSV-2	
Dist. from "FROM" Node / Offset dist	1176	mm.
Miscellaneous Weight	3500	Kgf
Offset from Element Centerline	5000	mm.
Element From Node	70	
Detail Type	Weight	
Detail ID	PSV-3	
Dist. from "FROM" Node / Offset dist	1176	mm.
Miscellaneous Weight	3500	Kgf
Offset from Element Centerline	5000	mm.
Element From Node	70	
Detail Type	Weight	
Detail ID	PSV-4	
Dist. from "FROM" Node / Offset dist	1176	mm.
Miscellaneous Weight	3500	Kgf
Offset from Element Centerline	5000	mm.
Element From Node	70	
Detail Type	For./Mom.	
Detail ID	N2- Force	
Dist. from "FROM" Node / Offset dist	1136	mm.
Force in X Direction	5889.3	Kgf
Force in Y Direction	-8833.9	Kgf
Force in Z Direction	5889.3	Kgf
Moment about X Axis	17308	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	17308	Kg-m.
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
	Skirt	...	2000	...	2000
	Skirt	...	4000	...	2000
	Skirt	...	5000	...	1000
	BOTTOM HEAD	...	5050	...	50
	SHELL-1	...	8050	...	3000
	SHELL-2	...	9450	...	1400
	TOP HEAD	...	9500	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
	Skirt	...	12	1	4502	...	1
	Skirt	...	12	1	4502	...	2
	Skirt	...	12	1	4502	...	3
	BOTTOM HEAD	3.6787	12	1	4506	117.3	4
	SHELL-1	3.5648	12	1	4506	117.3	5
	SHELL-2	3.5068	12	1	4506	117.3	6
	TOP HEAD	3.5068	12	1	4506	117.3	7

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
	Skirt	...	No Calc	No Calc	12	No Calc	1
	Skirt	...	No Calc	No Calc	12	No Calc	2
	Skirt	...	No Calc	No Calc	12	No Calc	3
	BOTTOM HEAD	3.4998	4.51	5.2	10	8.06725	4
	SHELL-1	3.4998	5.64	6.23	12	7.86301	5
	SHELL-2	3.4998	5.7	6.23	12	7.75105	6
	TOP HEAD	3.4998	4.68	5.2	10	7.73697	7

Minimum 3.5 5.2

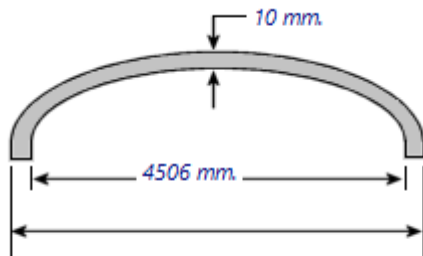
MAWP: 3.5 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: BOTTOM HEAD

Elliptical Head Design Information:					
Design Pressure	P	3.679 bars	Int. Design Temperature		110 °C
Material		SA-516 70	Ext. Design Temperature		110 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	4506.000 mm.	Actual Stress		108.4 N./mm ²
Spec. Min. Thk t _e , or s _a		10.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		229559.3 cm ²	SG of Contents		0.987
Element Volume		12775638.0 cm ³	Weight of Contents		12552.4 Kg _f
Empty Weight		2338.1 Kg _f	Operating Weight		14890.4 Kg _f
			Comp. All. Ope. Stress		59.6 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 56.325 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.163 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.68 \cdot 4508 \cdot 1) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 3.68)$$

$$= 7.0673 + 1.0000 = 8.0673 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.179 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 9) / (1 \cdot 4508 + 0.2 \cdot 9)$$

$$= 4.68 - 0.18 = 4.51 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 10) / (1 \cdot 4506 + 0.2 \cdot 10)$$

$$= 5.2 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.68 \cdot (1 \cdot 4508 + 0.2 \cdot 9)) / (2 \cdot 0.85 \cdot 9)$$

$$= 108.374 \text{ N./mm}^2$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (3.68 \cdot 2254) / (138 \cdot 0.85 - 0.6 \cdot 3.68) + 1$
 $= 8.083 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.070 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 0.85 \cdot 11) / (2254 + 0.6 \cdot 11)$
 $= 5.71 - 0.07 = 5.64 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (4508 / (2 \cdot 1128))^2) / 6$
 $= 0.999409$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.166 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10$, $t_r = 6.72$, $c = 1 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.63$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12$, $t_r = 6.74$, $c = 1 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	3.500 bars
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm.	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.521	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 56.325 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.163 mm.

EXTERNAL PRESSURE RESULTS for: BOTTOM HEAD

Elliptical Head From 40 to 50 Ext. Chart: CS-2 at 110 °C

Elastic Modulus from Chart: CS-2 at 110 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
9.000	4526.00	502.89	0.0002762	27.61

$MAEP = B / (K_0 \cdot Do / t) = 27.6 / (0.9 \cdot 503) = 0.61 \text{ bars}$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
8.286	4526.00	546.21	0.0002543	25.42

PV Elite 28 U1 Licensee: SL User
FileName : LP Flash Column Overhead Reflux Drum (153-D)_Rev Page 26 of 167
Element: BOTTOM HEAD Step: 4 1:54pm Sep 16, 2026



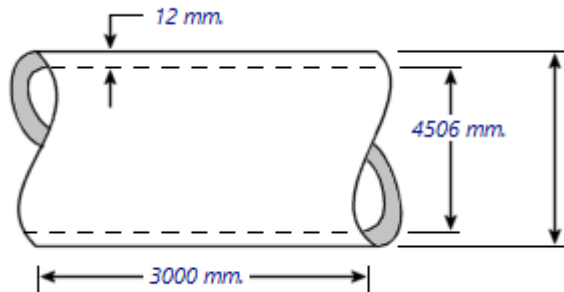
HEXAGON

$MAEP = B / (K_0 * D_o / t) = 25.4 / (0.9 * 546) = 0.52 \text{ bars}$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-1

Cylindrical Shell Design Information:					
Design Pressure	P	3.565 bars	Int. Design Temperature		110 °C
Material		SA-516 70	Ext. Design Temperature		110 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	4506.000 mm.	Actual Stress		86.2 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		426942.5 cm ²	SG of Contents		0.987
Element Volume		47848828.0 cm ³	Weight of Contents		9414.5 Kg _f
Empty Weight		3960.2 Kg _f	Operating Weight		13374.6 Kg _f
			Comp. All. Ope. Stress		72.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.56 \cdot 2254) / (138 \cdot 0.85 - 0.6 \cdot 3.56)$$

$$= 6.8630 + 1.0000 = 7.8630 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.065 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 11) / (2254 + 0.6 \cdot 11)$$

$$= 5.71 - 0.065 = 5.64 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 12) / (2253 + 0.6 \cdot 12)$$

$$= 6.23 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.56 \cdot (2254 + 0.6 \cdot 11)) / (0.85 \cdot 11)$$

$$= 86.194 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.266 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12, tr = 6.74, c = 1 mm., E* = 0.85



Thickness Ratio = $tr * E / (tg - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	3.500 bars
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm.	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.521	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: SHELL-1

Cylindrical Shell From 50 to Ring R1 Fr50 Ext. Chart: CS-2 at 110 °C

Elastic Modulus from Chart: CS-2 at 110 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	4530.00	925.50	411.82	0.2043	0.0008576	80.44

$$MAEP = (4*B)/(3*(Do/t)) = (4*80.4)/(3*412) = 2.6 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.697	4530.00	925.50	795.18	0.2043	0.0003085	30.84

$$MAEP = (4*B)/(3*(Do/t)) = (4*30.8)/(3*795) = 0.52 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	4530.00	4497.64	411.82	0.9929	0.0001599	15.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*16)/(3*412) = 0.52 \text{ bars}$$

Cylindrical Shell From Ring R1 Fr50 to 60 Ext. Chart: CS-2 at 110 °C

Elastic Modulus from Chart: CS-2 at 110 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	4530.00	3200.00	411.82	0.7064	0.0002268	22.68

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.7)/(3*412) = 0.73 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.552	4530.00	3200.00	474.22	0.7064	0.0001840	18.39

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.4)/(3*474) = 0.52 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
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PV Elite 28 U1 Licensee: SL User
FileName : LP Flash Column Overhead Reflux Drum (153-D)_Rev Page 29 of 167
Element: SHELL-1 Step: 5 1:54pm Sep 16,2026



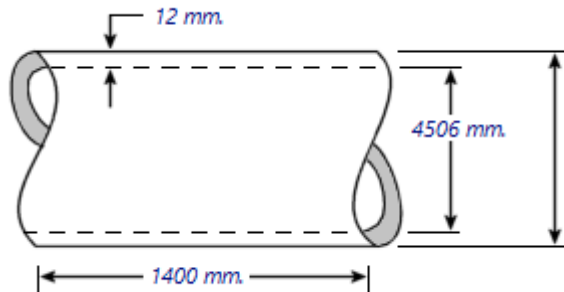
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	4530.00	4497.86	411.82	0.9929	0.0001598	15.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*16)/(3*412) = 0.52 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-2

Cylindrical Shell Design Information:					
Design Pressure	P	3.507 bars	Int. Design Temperature		110 °C
Material		SA-516 70	Ext. Design Temperature		110 °C
Length		1400.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	4506.000 mm.	Actual Stress		84.8 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		199239.8 cm ²	SG of Contents		0.000
Element Volume		22329454.0 cm ³	Weight of Contents		0.0 Kg
Empty Weight		1848.1 Kg	Operating Weight		1848.1 Kg
			Comp. All. Ope. Stress		72.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.51 \cdot 2254) / (138 \cdot 0.85 - 0.6 \cdot 3.51)$$

$$= 6.7511 + 1.0000 = 7.7511 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 11) / (2254 + 0.6 \cdot 11)$$

$$= 5.71 - 0.007 = 5.7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 12) / (2253 + 0.6 \cdot 12)$$

$$= 6.23 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.51 \cdot (2254 + 0.6 \cdot 11)) / (0.85 \cdot 11)$$

$$= 84.791 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.266 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12, tr = 6.74, c = 1 mm., E* = 0.85

Thickness Ratio = $tr * E / (tg - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	3.500 bars
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm.	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.521	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: SHELL-2

Cylindrical Shell From 60 to Ring R1 Fr60 Ext. Chart: CS-2 at 110 °C

Elastic Modulus from Chart: CS-2 at 110 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	4530.00	3200.00	411.82	0.7064	0.0002268	22.68

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.7)/(3*412) = 0.73 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.552	4530.00	3200.00	474.22	0.7064	0.0001840	18.39

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.4)/(3*474) = 0.52 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	4530.00	4497.86	411.82	0.9929	0.0001598	15.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*16)/(3*412) = 0.52 \text{ bars}$$

Cylindrical Shell From Ring R1 Fr60 to 70 Ext. Chart: CS-2 at 110 °C

Elastic Modulus from Chart: CS-2 at 110 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	4530.00	1125.50	411.82	0.2485	0.0006929	69.28

$$MAEP = (4*B)/(3*(Do/t)) = (4*69.3)/(3*412) = 2.24 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.147	4530.00	1125.50	736.94	0.2485	0.0002859	28.58

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.6)/(3*737) = 0.52 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
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PV Elite 28 U1 Licensee: SL User
FileName : LP Flash Column Overhead Reflux Drum (153-D)_Rev Page 32 of 167
Element: SHELL-2 Step: 6 1:54pm Sep 16,2026



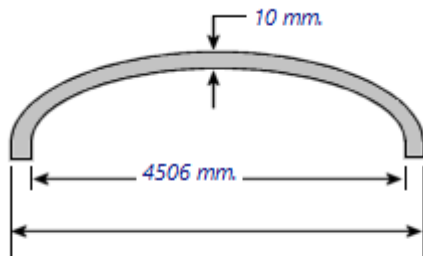
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	4530.00	4498.06	411.82	0.9929	0.0001598	15.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*16)/(3*412) = 0.52 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: TOP HEAD

Elliptical Head Design Information:					
Design Pressure	P	3.507 bars	Int. Design Temperature		110 °C
Material		SA-516 70	Ext. Design Temperature		110 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	4506.000 mm.	Actual Stress		103.3 N./mm ²
Spec. Min. Thk t _e , or s _a		10.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		229559.3 cm ²	SG of Contents		0.000
Element Volume		12775638.0 cm ³	Weight of Contents		0.0 Kg _f
Empty Weight		2338.1 Kg _f	Operating Weight		2338.1 Kg _f
			Comp. All. Ope. Stress		59.6 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 56.325 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.163 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.51 \cdot 4508 \cdot 1) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 3.51)$$

$$= 6.7370 + 1.0000 = 7.7370 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 9) / (1 \cdot 4508 + 0.2 \cdot 9)$$

$$= 4.68 - 0.007 = 4.68 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 10) / (1 \cdot 4506 + 0.2 \cdot 10)$$

$$= 5.2 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.51 \cdot (1 \cdot 4508 + 0.2 \cdot 9)) / (2 \cdot 0.85 \cdot 9)$$

$$= 103.311 \text{ N./mm}^2$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (3.51 \cdot 2254) / (138 \cdot 0.85 - 0.6 \cdot 3.51) + 1$
 $= 7.751 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.007 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 0.85 \cdot 11) / (2254 + 0.6 \cdot 11)$
 $= 5.71 - 0.007 = 5.7 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (4508 / (2 \cdot 1128))^2) / 6$
 $= 0.999409$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.166 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10$, $t_r = 6.72$, $c = 1 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.63$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12$, $t_r = 6.74$, $c = 1 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	3.500 bars
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm.	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.521	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 56.325 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.163 mm.

EXTERNAL PRESSURE RESULTS for: TOP HEAD

Elliptical Head From 70 to 80 Ext. Chart: CS-2 at 110 °C

Elastic Modulus from Chart: CS-2 at 110 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
9.000	4526.00	502.89	0.0002762	27.61

$MAEP = B / (K_0 \cdot Do / t) = 27.6 / (0.9 \cdot 503) = 0.61 \text{ bars}$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
8.286	4526.00	546.21	0.0002543	25.42

PV Elite 28 U1 Licensee: SL User
FileName : LP Flash Column Overhead Reflux Drum (153-D)_Rev Page 35 of 167
Element: TOP HEAD Step: 7 1:54pm Sep 16, 2026



HEXAGON

$MAEP = B / (K_0 * Do / t) = 25.4 / (0.9 * 546) = 0.52 \text{ bars}$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	4.550 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.323 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	3.850 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	3.500 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	179.4	234.0	0.767	6.76
SHELL-1	149.9	234.0	0.641	6.76
SHELL-2	149.9	234.0	0.641	6.76
TOP HEAD	179.4	234.0	0.767	6.76

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-4"	Nozzle	118.00	118.00	1.000
N3-4"	Pad	138.00	138.00	1.000
N1-42"	Nozzle	138.00	138.00	1.000
N1-42"	Pad	138.00	138.00	1.000
N4-2"	Nozzle	115.00	115.00	1.000
K1-2"	Nozzle	115.00	115.00	1.000
K2-2"	Nozzle	115.00	115.00	1.000
K4-2"	Nozzle	115.00	115.00	1.000
K5-2"	Nozzle	115.00	115.00	1.000
K3-2"	Nozzle	115.00	115.00	1.000
K6-2"	Nozzle	115.00	115.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
N2-42"	Nozzle	138.00	138.00	1.000
N2-42"	Pad	138.00	138.00	1.000
K7-2"	Nozzle	115.00	115.00	1.000
K8-2"	Nozzle	115.00	115.00	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring R1 Fr50	138.00	138.00	1.000
Ring R1 Fr60	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-4"	3.20	216.00	0.015
N1-42"	30.48	234.00	0.130
N4-2"	1.43	153.00	0.009
K1-2"	1.43	153.00	0.009
K2-2"	1.43	153.00	0.009
K4-2"	1.43	153.00	0.009
K5-2"	1.43	153.00	0.009
K3-2:	1.43	153.00	0.009
K6-2"	1.43	153.00	0.009
M1-24"	17.59	234.00	0.075
N2-42"	36.49	234.00	0.156
K7-2"	1.43	153.00	0.009
K8-2"	1.43	153.00	0.009

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	4.550 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.650 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	3.850 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	3.500 bars

Vertical Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	215.4	234.0	0.920	7.31
SHELL-1	174.0	234.0	0.744	7.20
SHELL-2	166.9	234.0	0.713	6.90
TOP HEAD	199.3	234.0	0.852	6.76

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-4"	Nozzle	118.00	118.00	1.000
N3-4"	Pad	138.00	138.00	1.000
N1-42"	Nozzle	138.00	138.00	1.000
N1-42"	Pad	138.00	138.00	1.000
N4-2"	Nozzle	115.00	115.00	1.000
K1-2"	Nozzle	115.00	115.00	1.000
K2-2"	Nozzle	115.00	115.00	1.000
K4-2"	Nozzle	115.00	115.00	1.000
K5-2"	Nozzle	115.00	115.00	1.000
K3-2"	Nozzle	115.00	115.00	1.000
K6-2"	Nozzle	115.00	115.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
N2-42"	Nozzle	138.00	138.00	1.000
N2-42"	Pad	138.00	138.00	1.000
K7-2"	Nozzle	115.00	115.00	1.000
K8-2"	Nozzle	115.00	115.00	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring R1 Fr50	138.00	138.00	1.000
Ring R1 Fr60	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-4"	3.46	216.00	0.016
N1-42"	32.42	234.00	0.139
N4-2"	1.52	153.00	0.010
K1-2"	1.52	153.00	0.010
K2-2"	1.52	153.00	0.010
K4-2"	1.52	153.00	0.010
K5-2"	1.52	153.00	0.010
K3-2:	1.52	153.00	0.010
K6-2"	1.52	153.00	0.010
M1-24"	18.71	234.00	0.080
N2-42"	36.49	234.00	0.156
K7-2"	1.43	153.00	0.009
K8-2"	1.43	153.00	0.009

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

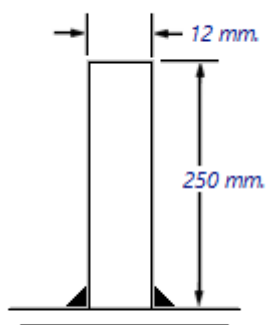
MDMT Calculations for: Ring R1 Fr50, Shell to Ring Junction, min(Curve:B, Curve:D)

Govrn. thk, $t_g = 12$, $t_r = 6.74$, $c = 1$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Stiffening Ring Calculations for: Ring R1 Fr50, SA-516 70 , Bar Ring: 250.0 x 12.0 mm.

Material treatment options:



Effective Length of Shell: 246 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	27.010	5.5000	148.557
Ring :	30.000	135.9999	4079.995
Total:	57.010		4228.553

Centroid of Ring plus Shell: 74 mm.

	Inertia	Distance	A*Dist ²
Shell:	2.724	68.6716	127375.406
Ring :	1562.496	-61.8283	114682.078
Total:	1565.219		242057.484

Available Moment of Inertia, Ring + Shell: 0.399E+04 cm⁴

Required Stress in Ring plus Shell Breq 14.11 N./mm²
 Required Strain in Ring plus Shell Areq 0.0001410

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4530^2 \cdot 2063(11 + 30/2063)0.000141) / 10.9$$

$$= 682 \text{ cm}^4$$

MDMT Calculations for: Ring R1 Fr60, Shell to Ring Junction, min(Curve:B, Curve:D)

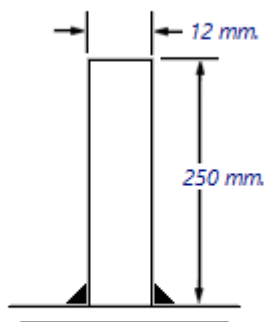
Govrn. thk, $t_g = 12$, $t_r = 6.74$, $c = 1$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Stiffening Ring Calculations for: Ring R1 Fr60, SA-516 70 , Bar Ring: 250.0 x 12.0 mm.

Material treatment options:



Effective Length of Shell: 246 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	27.010	5.5000	148.557
Ring :	30.000	135.9999	4079.995
Total:	57.010		4228.553

Centroid of Ring plus Shell: 74 mm.

	Inertia	Distance	A*Dist ²
Shell:	2.724	68.6716	127375.406
Ring :	1562.496	-61.8283	114682.078
Total:	1565.219		242057.484

Available Moment of Inertia, Ring + Shell: 0.399E+04 cm⁴

Required Stress in Ring plus Shell Breq 14.18 N./mm²
 Required Strain in Ring plus Shell Areq 0.0001420

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4530^2 \cdot 2163(11 + 30/2163)0.000142) / 10.9$$

$$= 716 \text{ cm}^4$$

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	5000	...	...	No Calc	No Calc
20	30	5000	...	...	No Calc	No Calc
30	40	5000	...	...	No Calc	No Calc
40	50	No Calc	4526	9	0.00027618	27.612
50	Ring	925.5	4530	11	0.00085764	80.4383
Ring	60	3200	4530	11	0.0002268	22.6751
60	Ring	3200	4530	11	0.0002268	22.6751
Ring	70	1125.5	4530	11	0.00069292	69.2768
70	80	No Calc	4526	9	0.00027618	27.612



External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	10	9.28625	0.51708	0.61
50	Ring	12	6.69686	0.51708	2.6
Ring	60	12	10.5524	0.51708	0.73
60	Ring	12	10.5524	0.51708	0.73
Ring	70	12	7.14708	0.51708	2.24
70	80	10	9.28625	0.51708	0.61

Minimum

0.61

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	5000	No Calc	No Calc	No Calc
20	30	5000	No Calc	No Calc	No Calc
30	40	5000	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	Ring	925.5	4497.64	No Calc	No Calc
Ring	60	3200	4497.86	681.956	3985.79
60	Ring	3200	4497.86	No Calc	No Calc
Ring	70	1125.5	4498.06	716.2	3985.79
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	4434.56	...	4215.28	...	310.419
20	30	2637.89	...	2418.61	...	184.652
30	40	1318.88	...	1209.24	...	92.3215
40	50	2338.07	12775638	2143.23	12792301	163.665
50	60	3960.16	47848828	3630.97	47891312	277.211
60	70	1848.06	22329454	1694.44	22349280	129.364
70	80	2338.07	12775638	2143.23	12792301	163.665
Total		18875	95729560.00	17455	95825200.00	1321

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
40	Plat	5334.03	...	...	0.27881E-04	PLAT:[1 OF 1]
40	Liqd	12552.4	...	-563.25	...	Liquid: 20
40	Lini	600.124	...	-538.25	...	SS 304L CLAD
40	Nozl	103.618	...	-1296.51	...	N3-4"
40	Wght	25	...	-1125	...	N3-Anti Swirl B
50	Liqd	9414.48	...	300	...	Liquid: 30
50	Lini	1019.71	...	1500	...	Lining
50	Ring	373.605	...	500	...	Ring R1 Fr50
50	Nozl	1714.3	2428.01	2000	...	N1-42"
50	Nozl	10.8632	-1649.68	850	1649.68	N4-2"
50	Nozl	10.3488	-2333	1800	...	K1-2"
50	Nozl	10.3488	-2333	250	...	K2-2"
50	Nozl	10.3488	1649.68	250	-1649.68	K4-2"
50	Nozl	10.3488	1649.68	850	1649.68	K5-2"
50	Nozl	10.3488	-2333	850	...	K3-2:
50	Nozl	10.3488	-1649.68	250	1649.68	K6-2"
50	Nozl	441.569	...	2000	-2428.01	M1-24"
50	Wght	50	...	2000	-2614	M1 Davit
50	Wght	1144	...	1500	...	Clip Weight
60	Lini	475.866	...	700	...	SS 304 L CLAD
60	Ring	373.605	...	700	...	Ring R1 Fr60
60	Wght	350	...	700	...	Demister
60	Wght	121	...	700	...	Demisting Suppo
70	Plat	18223	...	1500	...	SQ PLT
70	Lini	600.124	...	588.25	...	SS 304L CLAD
70	Nozl	484.55	...	1359	...	N2-42"
70	Nozl	28.2587	-1500	1260.54	...	K7-2"
70	Nozl	28.2587	1500	1260.54	...	K8-2"
70	Wght	3500	3535.53	1176	-3535.53	PSV Weigt-1
70	Wght	3500	3535.53	1176	3535.53	PSV-2
70	Wght	3500	-3535.53	1176	3535.53	PSV-3
70	Wght	3500	-3535.53	1176	-3535.53	PSV-4
70	Forc	...	...	1136	...	N2- Force

Total Weight of Each Detail Type:

Platforms	23557.0
Liquid	21966.8
Cladding & Overlay	2695.8
Stiffeners	747.2
Nozzles	2873.5
Weights	15690.0
Sum of the Detail Weights	67530.4 kg.

Weight Summation Results: (kg.)



	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	20197.0	20197.0	20197.0	20197.0	20197.0	20197.0
Cladding	2695.8	2695.8	2695.8	2695.8	2695.8	2695.8
Stif. Rings	747.2	747.2	747.2	747.2	747.2	747.2
Nozzles	2873.5	2873.5	2873.5	2873.5	2873.5	2873.5
Wld Weights	1340.0	1340.0	1340.0	1340.0	1340.0	1340.0
Platforms	...	...	...	23557.0	23557.0	23557.0
Empty Weights	...	...	...	14350.0	14350.0	...
Test Weights	...	14350.0	...	...	...	...
Ope Weights	...	...	...	...	...	14350.0
Ope. Liquid	...	...	...	...	...	21966.8
Test Liquid	...	95671.1	...	...	...	...
Totals	27853.5	137874.7	27853.5	65760.5	65760.5	87727.4

Field Installation Options:

Platforms installed before lifting.

* Lining installed after lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	27853.5 kg.
Shop Test Wt.	- Fabricated weight + water (full)	137874.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	27853.5 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	65760.5 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	65760.5 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	87727.4 kg.
Field Test Wt.	- Empty weight + test medium (full)	161095.2 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	44957.8 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	105577.0 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	45532.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	284377	282869
20	30	284377	282869
30	40	142188	141434
40	50	229559	227171
50	60	426942	424681
60	70	199240	198184
70	80	229559	227171

Total	1796243.4	1784378.9 cm ²
Total	1933.5	1920.7 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	4744.98	4744.98	4744.98	...	4744.98
20	30	2822.55	2822.55	2822.55	...	2822.55
30	40	1411.2	1411.2	1411.2	...	1411.2
40	50	2630.36	21116.9	21332.3	...	8564.51



50	60	8033.8	18468	56873.1	4089.13	9053.51
60	70	2822.03	3297.9	25613.7	...	3297.9
70	80	17042.8	35865.9	48633.7	...	35865.9

Empty Support Force + the Sum of the Y forces 48341.61 Kgf
 Operating Support Force + the Sum of the Y forces 96561.27 Kgf
 Hydro Support Force + the Sum of the Y forces 170265.52 Kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	65760.5	87727.4	161432
20	30	61015.6	82982.4	156687
30	40	58193	80159.9	153864
40	50	56781.8	78748.7	152453
50	60	48217.3	57631.8	131121
60	70	39163.8	39163.8	74247.4
70	80	35865.9	35865.9	48633.7

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	4089.13	4089.13	4089.13
20	30	4089.13	4089.13	4089.13
30	40	4089.13	4089.13	4089.13
40	50	4089.13	4089.13	4089.13
50	60	4089.13	4089.13	4089.13
60	70	...	...	...
70	80	...	...	...

Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N3-4"	46.38	51.00	110	300	GR 1.1	38.210	31.361	46.380	31.361
N1-42"	17.32	19.60	110	150	GR 1.1	8.417	11.872	17.320	17.320
N4-2"	68.50	82.30	110	600	GR 2.3	76.295	26.457	68.501	26.457
K1-2"	34.28	41.10	110	300	GR 2.3	...	...	...	...
K2-2"	34.28	41.10	110	300	GR 2.3	...	...	...	...
K4-2"	34.28	41.10	110	300	GR 2.3	...	...	...	...
K5-2"	34.28	41.10	110	300	GR 2.3	...	...	...	...
K3-2:	34.28	41.10	110	300	GR 2.3	...	...	...	...
K6-2"	34.28	41.10	110	300	GR 2.3	...	...	...	...
M1-24"	17.32	19.60	110	150	GR 1.1	...	...	...	...
N2-42"	17.32	19.60	110	150	GR 1.1	8.417	11.872	17.320	17.320
K7-2"	34.28	41.10	110	300	GR 2.3	...	...	...	...
K8-2"	34.28	41.10	110	300	GR 2.3	...	...	...	...
Min Rating	11.872	19.600, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N3-4"	138.176	0.50	202	0	3.500	31.361	Pass
N1-42"	1097.026	0.17	22252	0	3.500	11.872	Pass
N4-2"	77.851	0.50	72	0	3.500	26.457	Pass
N2-42"	1097.026	0.17	22252	0	3.500	11.872	Pass

Evaluating Flange on Nozzle [N3-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(202) + 4 \cdot \text{abs}(0)138.176 \leq \pi \cdot 138.176^3 [(46.380 - 3.500) + .500 \cdot 46.380]$$

$$= 0.32294E+04 \leq 0.55842E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(46.380, 0.500 \cdot 46.380 + 46.380 - 3229.4 / (\pi \cdot 138.176^3))$$

$$= \min(46.380, 31.361)$$

$$= 31.361 \text{ bars}$$

Evaluating Flange on Nozzle [N1-42"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(22252) + 4 \cdot \text{abs}(0)1097.026 \leq \pi \cdot 1097.026^3 [(17.320 - 3.500) + .171 \cdot 17.320]$$

$$= 0.35603E+06 \leq 0.71013E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-42" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(17.320, 0.171 \cdot 17.320 + 17.320 - 356025.8 / (\pi \cdot 1097.026^3))$$

$$= \min(17.320, 11.872)$$

$$= 11.872 \text{ bars}$$

Evaluating Flange on Nozzle [N4-2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(72) + 4 \cdot \text{abs}(0)77.851 \leq \pi \cdot 77.851^3 [(68.501 - 3.500) + .500 \cdot 68.501]$$

= 0.11533E+04 ≤ 0.15003E+04 Check Passes

Maximum Pressure for Flange on Nozzle N4-2" [Pmax]:
= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
= min(68.501, 0.500 • 68.501 + 68.501 - 1153.3/(π • 77.851³))
= min(68.501, 26.457)
= 26.457 bars

Evaluating Flange on Nozzle [N2-42"]:
= 16•ME + 4•FE•G ≤ π•G³[(PR - PD) + FM • PR]
= 16(22252)+4•abs(0)1097.026 ≤ π•1097.026³[(17.320-3.500)+.171•17.320]
= 0.35603E+06 ≤ 0.71013E+06 Check Passes

Maximum Pressure for Flange on Nozzle N2-42" [Pmax]:
= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
= min(17.320, 0.171 • 17.320 + 17.320 - 356025.8/(π • 1097.026³))
= min(17.320, 11.872)
= 11.872 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 10.4969 Hz.

The Natural Frequency for the Vessel (Ope...) is 9.81955 Hz.

The Natural Frequency for the Vessel (Filled) is 7.12197 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	8328.66	33938.4

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	8328.66	33938.4

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
70	1136.00	5889.	-8834.	5889.	17308.		17308.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.22 \cdot 0.93) / (1 + 1.7 \cdot 3.4 \cdot 0.22)))$
 $= \min(0.85, 0.89)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	9.820 Hz
Natural Frequency of Vessel (Empty)	10.497 Hz
Natural Frequency of Vessel (Test)	7.122 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.524
Structure Height to Diameter ratio	2.416
Height to top of Structure	10636.500 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm.}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000 + 300 = 1300 \text{ mm.}$
 $= 4.27 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because $z (4.27 \text{ ft.}) < 15 \text{ ft.}$
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[Kzt]$:



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$$

$$= 16 \text{ psf } [78.1] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16 \cdot 0.85 \cdot 0.52 \cdot 127$$

$$= 902 \text{ lbs. } [409] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Skirt	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Skirt	3300.0	0.000	0.000	0.000	0.849	1.000	78.120
Skirt	4800.0	0.000	0.000	0.000	0.858	1.000	78.120
BOTTOM HEAD	5325.0	0.000	0.000	0.000	0.877	1.000	78.120
SHELL-1	6850.0	0.000	0.000	0.000	0.924	1.000	78.120
SHELL-2	9050.0	0.000	0.000	0.000	0.980	1.000	78.120
TOP HEAD	10255.4	0.000	0.000	0.000	1.006	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of (D/L²)(10⁴)
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kg
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.32037E-06 (High); 0.32037E-06 < 0.40047E-06 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0/9500 = 0

:: [D / L²] • 10⁴ < 8 (English Units, ft.)
 :: [14.9/31.2²] • 10⁴ = 153 [Geometry Violation]

Compute the vibration possibility. If Vp > 0.40047E-06 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 86207 / (9500 \cdot 4530^2)$$

$$= 0.44226E-06$$

Since Vp is > 0.40047E-06 no further vibration analysis is required !

Platform Load Calculations



ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	30000.00	5300.00	78.12	234.36	1.00
SQ PLT	100000.00	11250.00	78.12	781.20	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
N3-Anti Swirl	0.00	4175.00	78.12	0.00
M1 Davit	0.00	7350.00	78.12	0.00
Clip Weight	0.00	6850.00	78.12	0.00
Demister	0.00	9050.00	78.12	0.00
Demisting Sup	0.00	9050.00	78.12	0.00
PSV Weigt-1	0.00	10926.00	78.12	0.00
PSV-2	0.00	10926.00	78.12	0.00
PSV-3	0.00	10926.00	78.12	0.00
PSV-4	0.00	10926.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 9.81955 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1300	5883.8	117676	78.12	450.051
20	30	3300	5883.8	117676	78.12	450.051
30	40	4800	5883.8	58838	78.12	225.025
40	50	5325	5883.8	2941.9	78.12	267.837
50	60	6850	5889	176670	78.12	675.673
60	70	9050	5889	82446	78.12	315.314
70	80	10255.4	5883.8	55461	78.12	1154.77

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.3 * 0.15 = 0.19 \\
 SM1 &= Fv * S1 = 1.5 * 0.065 = 0.097 \\
 SDS &= 2/3 * Sms = 2/3 * 0.19 = 0.13 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.097 = 0.065
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(34.90.75)$
 $= 0.287$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/9.82)$
 $= 0.102$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.29, 0.1)$ per 12.8.2
 $= 0.102$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R / I)$
 $= 0.13 / (1.25 / 1.25)$
 $= 0.127$

Maximum value of C_s per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R / I_e))$
 $= 0.065 / (0.1(1.25 / 1.25))$
 $= 0.638$

Minimum value of C_s per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot SDS \cdot I_e, 0.030)$
 $= \max(0.044 \cdot 0.13 \cdot 1.25, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= C_s \cdot W$ (Equation 12.8-1):
 $= 0.13 \cdot 87727$
 $= 11176.469$ Kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.13 \cdot 87727$



= 2235.3 Kgf

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 9.81955 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	1000	4744.98	86.1854	87.9419	-6598.59
20	30	3000	2822.55	153.802	156.936	-6719.49
30	40	4500	1411.2	115.345	117.696	-6791.41
40	50	5025	21116.9	1927.37	244.969	-6827.36
50	60	6550	18468	2197.15	975.267	-7365.42
60	70	8750	3297.9	524.136	457.647	-7835.99
70	80	9475	35865.9	6172.48	2992.83	-7920.02



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm.	Cumulative Shr Wind Cas Kg-f	Cumulative Shr Eqk Cas Kg-f	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	8328.66	8328.66	112646	112646
20	30	3000	8328.66	8328.66	95988.1	95988.1
30	40	4500	8328.66	8328.66	79330.5	79330.5
40	50	5025	8328.66	8328.66	71001.6	71001.6
50	60	6550	8328.66	8328.66	70585.2	70585.2
60	70	8750	8328.66	8328.66	45598.7	45598.7
70	80	9475	8328.66	8328.66	33938.4	33938.4

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	3538.72	11176.5	22840	91180.7
20	30	3000	3088.67	11090.3	16212.5	68913.5
30	40	4500	2638.62	10936.5	10485.1	46886.3
40	50	5025	2413.59	10821.1	7958.95	36007.3
50	60	6550	2145.76	8893.77	7844.97	35514.4
60	70	8750	1470.08	6696.62	2421.1	12128.4
70	80	9475	1154.77	6172.48	583.67	3119.84

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	8328.66	0.006426	0.025055	0.24405E-04
20	30	8328.66	0.054912	0.095023	0.44913E-04
30	40	8328.66	0.11862	0.14441	0.53706E-04
40	50	8328.66	0.14576	0.14711	0.54234E-04
50	60	8328.66	0.23757	0.34397	0.75483E-04
60	70	8328.66	0.39811	0.45463	0.82272E-04
70	80	8328.66	0.45669	0.45875	0.82272E-04

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 47.500 Actual deflection : 0.459 mm.

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	3538.72	0.001295	0.0049117	0.46478E-05
20	30	3088.67	0.010468	0.017607	0.78195E-05
30	40	2638.62	0.021667	0.026002	0.89190E-05
40	50	2413.59	0.026225	0.026449	0.89792E-05
50	60	2145.76	0.040872	0.056675	0.10834E-04
60	70	1470.08	0.064323	0.072059	0.11089E-04
70	80	1154.77	0.072336	0.072613	0.11089E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	1037.17	6482.32
20	30	1037.17	6482.32
30	40	1037.17	6482.32
40	50	1037.17	6482.32
50	60	1038.09	6488.05
60	70	1038.09	6488.05
70	80	1037.17	6482.32

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 47.500 Actual deflection : 0.073 mm.

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 47.500 Actual deflection : 0.531 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	1701.75	1560.28	191533904	175690384
20	30	1701.75	1560.28	191533904	175690384
30	40	1701.75	1560.28	191533904	175690384
40	50	1418.74	1277.16	159822592	143936896
50	60	1703.26	1561.68	191873968	176002304
60	70	1703.26	1561.68	191873968	176002304
70	80	1418.74	1277.16	159822592	143936896

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	95.7583	188.997	-72.8957	-72.8957
20	30	95.7583	188.997	-72.8957	-72.8957
30	40	82.8	140.4	-72.8957	-72.8957
40	50	140.76	238.68	-59.6419	-59.6419
50	60	140.76	238.68	-72.8313	-72.8313
60	70	140.76	238.68	-72.8313	-72.8313
70	80	140.76	238.68	-59.6419	-59.6419

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	43.8453	-6.51409	83.1384
50	60	35.8607	-5.33673	67.9981
60	70	35.8607	-5.33673	67.9981
70	80	43.8453	-6.51409	83.1384

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-4.13324	-5.51392	-10.1464	0.22825	0.22825
20	30	-3.835	-5.21568	-9.84821	0.22825	0.22825
30	40	-3.6576	-5.03828	-9.67081	0.22825	0.22825
40	50	-4.36006	-4.36006	-4.36006	0.2786	0.2786
50	60	-3.02789	-3.02789	-3.02789	0.22784	0.22784
60	70	-2.45936	-2.45936	-2.45936	...	...
70	80	-2.75401	-2.75401	-2.75401	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	0.22825	1.27488	5.0895	0.63744	...
20	30	0.22825	0.90495	3.8466	0.45247	...
30	40	0.22825	0.58525	2.61709	0.29263	...
40	50	0.2786	0.54226	2.45323	0.27113	...
50	60	0.22784	0.43711	1.97882	0.21856	...
60	70	...	0.1349	0.67578	0.06745	...
70	80	...	0.039766	0.21256	0.019883	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	2.37267
20	30	...	...	...	1.81568
30	40	...	...	...	1.27235
40	50	...	...	...	1.2308
50	60	...	...	...	0.99389
60	70	...	...	...	0.3356
70	80	...	...	...	0.10306

Longitudinal Stresses due to Applied Axial Forces:



From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	-0.55523	-0.41474
20	30	-0.55523	-0.42234
30	40	-0.55523	-0.42686
40	50	-0.67832	-0.52425
50	60	-0.55474	-0.46252
60	70	-0.55474	-0.49207
70	80	-0.67832	-0.60815

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	6.28763	6.28763	5.76752	5.76752
20	30	5.35784	5.35784	4.91464	4.91464
30	40	4.42805	4.42805	4.06176	4.06176
40	50	4.83745	4.83745	4.35663	4.35663
50	60	3.93292	3.93292	3.60759	3.60759
60	70	2.54071	2.54071	2.33054	2.33054
70	80	2.31227	2.31227	2.08244	2.08244

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	3.10	95.76	-12.48	72.90	0.0324	0.1712
20	2.10	95.76	-10.88	72.90	0.0219	0.1493
30	1.03	82.80	-9.45	72.90	0.0124	0.1297
40	0.62	140.76	-10.70	59.64	0.0044	0.1793
50	1.02	140.76	-8.18	72.83	0.0072	0.1123
60		140.76	-5.69	72.83		0.0781
70		140.76	-5.78	59.64		0.0970

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.34	95.76	-13.44	72.90	0.0453	0.1843
20	3.14	95.76	-11.66	72.90	0.0328	0.1599
30	1.84	82.80	-10.01	72.90	0.0223	0.1374
40	1.46	140.76	-11.23	59.64	0.0104	0.1883
50	1.66	140.76	-8.65	72.83	0.0118	0.1187
60		140.76	-5.83	72.83		0.0800
70		140.76	-5.78	59.64		0.0969

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10	1.72	95.76	-13.86	72.90	0.0180	0.1901
20	0.72	95.76	-12.26	72.90	0.0075	0.1682
30		82.80	-10.84	72.90		0.1486
40	0.62	140.76	-10.70	59.64	0.0044	0.1793
50	1.02	140.76	-8.18	72.83	0.0072	0.1123
60		140.76	-5.69	72.83		0.0781
70		140.76	-5.78	59.64		0.0970

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.68	95.76	-17.53	72.90	0.0593	0.2405
20	3.79	95.76	-15.07	72.90	0.0396	0.2067
30	1.81	82.80	-12.74	72.90	0.0218	0.1748
40	2.68	140.76	-12.45	59.64	0.0191	0.2088
50	2.65	140.76	-9.63	72.83	0.0188	0.1322
60	0.27	140.76	-6.17	72.83	0.0019	0.0847
70		140.76	-5.89	59.64		0.0987

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-11.01	72.90		0.1511
20		189.00	-10.53	72.90		0.1444
30		140.40	-10.19	72.90		0.1398
40		238.68	-4.91	59.64		0.0823
50		238.68	-3.47	72.83		0.0477
60		238.68	-2.53	72.83		0.0347
70		238.68	-2.77	59.64		0.0465

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-10.37	72.90		0.1423
20		189.00	-10.08	72.90		0.1382
30		140.40	-9.90	72.90		0.1358
40		238.68	-4.64	59.64		0.0778
50		238.68	-3.26	72.83		0.0447
60		238.68	-2.46	72.83		0.0338
70		238.68	-2.75	59.64		0.0462

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.72	95.76	-13.86	72.90	0.0180	0.1901
20	0.72	95.76	-12.26	72.90	0.0075	0.1682
30		82.80	-10.84	72.90		0.1486
40	44.47	140.76		59.64	0.3159	
50	36.88	140.76		72.83	0.2620	
60	35.52	140.76		72.83	0.2524	
70	42.77	140.76		59.64	0.3038	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.68	95.76	-17.53	72.90	0.0593	0.2405
20	3.79	95.76	-15.07	72.90	0.0396	0.2067
30	1.81	82.80	-12.74	72.90	0.0218	0.1748
40	46.53	140.76		59.64	0.3306	
50	38.51	140.76		72.83	0.2736	
60	36.13	140.76		72.83	0.2566	
70	43.01	140.76		59.64	0.3055	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10	1.72	95.76	-13.86	72.90	0.0180	0.1901
20	0.72	95.76	-12.26	72.90	0.0075	0.1682
30		82.80	-10.84	72.90		0.1486
40		140.76	-17.21	59.64		0.2886
50		140.76	-13.52	72.83		0.1856
60		140.76	-11.03	72.83		0.1514
70		140.76	-12.30	59.64		0.2062

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.68	95.76	-17.53	72.90	0.0593	0.2405
20	3.79	95.76	-15.07	72.90	0.0396	0.2067
30	1.81	82.80	-12.74	72.90	0.0218	0.1748
40		140.76	-18.97	59.64		0.3180
50		140.76	-14.97	72.83		0.2055
60		140.76	-11.50	72.83		0.1580
70		140.76	-12.40	59.64		0.2079

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-11.01	72.90		0.1511
20		189.00	-10.53	72.90		0.1444
30		140.40	-10.19	72.90		0.1398
40	79.33	238.68		59.64	0.3324	
50	65.42	238.68		72.83	0.2741	
60	65.61	238.68		72.83	0.2749	
70	80.40	238.68		59.64	0.3369	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-10.37	72.90		0.1423
20		189.00	-10.08	72.90		0.1382
30		140.40	-9.90	72.90		0.1358
40	79.06	238.68		59.64	0.3312	
50	65.20	238.68		72.83	0.2732	
60	65.54	238.68		72.83	0.2746	
70	80.38	238.68		59.64	0.3368	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.36	72.90		0.0598
20		95.76	-4.06	72.90		0.0557
30		82.80	-3.89	72.90		0.0533
40	39.76	140.76		59.64	0.2825	
50	33.06	140.76		72.83	0.2349	
60	33.40	140.76		72.83	0.2373	
70	41.09	140.76		59.64	0.2919	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-5.06	72.90		0.0694
20		95.76	-4.78	72.90		0.0656
30		82.80	-4.62	72.90		0.0634
40	39.92	140.76		59.64	0.2836	
50	33.08	140.76		72.83	0.2350	
60	33.61	140.76		72.83	0.2387	
70	41.37	140.76		59.64	0.2939	



Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-5.74	72.90		0.0788
20		95.76	-5.44	72.90		0.0747
30		82.80	-5.27	72.90		0.0722
40	39.76	140.76		59.64	0.2825	
50	33.06	140.76		72.83	0.2349	
60	33.40	140.76		72.83	0.2373	
70	41.09	140.76		59.64	0.2919	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.36	72.90		0.0598
20		95.76	-4.06	72.90		0.0557
30		82.80	-3.89	72.90		0.0533
40	39.76	140.76		59.64	0.2825	
50	33.06	140.76		72.83	0.2349	
60	33.40	140.76		72.83	0.2373	
70	41.09	140.76		59.64	0.2919	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-5.74	72.90		0.0788
20		95.76	-5.44	72.90		0.0747
30		82.80	-5.27	72.90		0.0722
40		140.76	-4.64	59.64		0.0778
50		140.76	-3.26	72.83		0.0447
60		140.76	-2.46	72.83		0.0338
70		140.76	-2.75	59.64		0.0462

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.59	79.80	-12.44	60.75	0.0074	0.2049
20		79.80	-11.22	60.75		0.1848
30		69.00	-10.12	60.75		0.1666
40	44.08	117.30		49.70	0.3758	
50	36.53	117.30		60.69	0.3114	
60	35.45	117.30		60.69	0.3022	
70	42.80	117.30		49.70	0.3648	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.59	79.80	-12.44	60.75	0.0074	0.2049
20		79.80	-11.22	60.75		0.1848
30		69.00	-10.12	60.75		0.1666
40		117.30	-16.51	49.70		0.3323
50		117.30	-12.99	60.69		0.2140
60		117.30	-10.83	60.69		0.1784
70		117.30	-12.19	49.70		0.2452

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-5.74	72.90		0.0788
20		95.76	-5.44	72.90		0.0747
30		82.80	-5.27	72.90		0.0722
40	39.76	140.76		59.64	0.2825	
50	33.06	140.76		72.83	0.2349	
60	33.40	140.76		72.83	0.2373	



70 41.09 140.76 59.64 0.2919

Absolute Maximum of the all of the Stress Ratio's 0.3758

Governing Element: BOTTOM HEAD
 Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt	...	-5.9	0.2	6.3	0.6	+ Tens
Skirt	...	-5.9	-0.2	-6.3	-12.4	- Comp
Skirt	...	-5.6	0.2	5.4	...	+ Tens
Skirt	...	-5.6	-0.2	-5.4	-11.2	- Comp
Skirt	...	-5.5	0.2	4.4	...	+ Tens
Skirt	...	-5.5	-0.2	-4.4	-10.1	- Comp
BOTTOM HEAD	43.8	-4.9	0.3	4.8	44.1	+ Tens
BOTTOM HEAD	43.8	-4.9	-0.3	-4.8	...	- Comp
SHELL-1	35.9	-3.5	0.2	3.9	36.5	+ Tens
SHELL-1	35.9	-3.5	-0.2	-3.9	...	- Comp
SHELL-2	35.9	-3.0	...	2.5	35.4	+ Tens
SHELL-2	35.9	-3.0	...	-2.5	...	- Comp
TOP HEAD	43.8	-3.4	...	2.3	42.8	+ Tens
TOP HEAD	43.8	-3.4	...	-2.3	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt

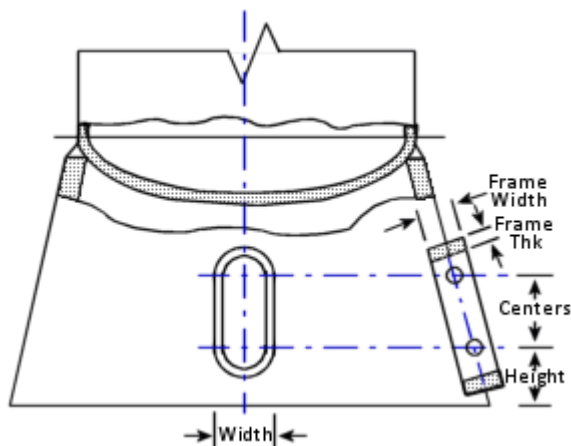
Total vessel (empty) weight	Fg	57369.195 Kgf
Head Weight below section 2-2	DeltaFg	8564.509 Kgf
Weight of vessel contents (liquid)	Ff	21966.844 Kgf
Vessel Gross Weight	Fg + Ff	79336.039 Kgf
Skirt Mean Diameter	Dz	4515.000 mm.
Skirt Analysis Thickness (Corroded)	ez	11.000 mm.
Skirt Height	H	1000.000 mm.
Additional Axial Force at section 1-1	F1	-8833.880 Kgf
Moment at section 1-1	M1	107008.953 Kg-m.
Applied Moment - Bottom of Skirt	Mb	126216.812 Kg-m.
Weight at Bottom of Skirt	W	80159.852 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	4517.000 mm.
Head Corroded Thickness	eb	9.000 mm.

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 1128 \text{ mm.} \\
 K &= D_i / (2 * h_i) = 4508 / (2 * 1128) = 2 \\
 R &= D_i (0.44 * K + 0.02) = 4508 * (0.44 * 2 + 0.02) = 4055 \text{ mm.} \\
 r &= D_i ((0.5 / K) - 0.08) = 4508 * ((0.5 / 2) - 0.08) = 767 \text{ mm.}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	39758080.000 cm ⁴
Moment of Inertia	Iyy =	39758080.000 cm ⁴
Section Modulus	W =	176115696.000 mm. ³
Section Metal Area	A =	1560.272 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

$$\begin{aligned}
 &\text{Radius of Gyration of Skirt at 4-4 [rg]:} \\
 &= \sqrt{\min(I_{xx}, I_{yy}) / \text{Area}} \\
 &= \sqrt{\min(39758080, 39758080) / 1560} \\
 &= 1596 \text{ mm.}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 1596/1000$$

$$= 1.596$$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 107009 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:
 Downwind Side:
 $= -M / Z - W / A$
 $= -107009/704462784 - 80160/1560$
 $= -6.53 \text{ N./mm}^2$

Upwind Side:
 $= +(M) / Z - W / A$
 $= +107009/704462784 - 80160/1560$
 $= -3.55 \text{ N./mm}^2$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:
 $= F1 - Fg - Ff + 4 \cdot M1 / Dz$
 $= -8834 - 57369 - 21967 + 4 \cdot 107009/4515$
 $= 6631 \text{ Kgf}$

Force at the joint, 16.12-2 [Fzq]:
 $= F1 - Fg - Ff - 4 \cdot M1 / Dz$
 $= -8834 - 57369 - 21967 - 4 \cdot 107009/4515$
 $= -182971 \text{ Kgf}$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:
 $= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (6631 + 8565 + 21967) / (\pi \cdot 4517 \cdot 9) + 3.5 \cdot 4517 / (4 \cdot 9)$
 $= 46.77 \text{ N./mm}^2$

Stress [σ_{1qm}]:
 $= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-182971 + 8565 + 21967) / (\pi \cdot 4517 \cdot 9) + 3.5 \cdot 4517 / (4 \cdot 9)$
 $= 32.21 \text{ N./mm}^2$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:
 $= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (21967 + 8565) / (\pi \cdot 4517 \cdot 9) + 3.5 \cdot 4517 / (4 \cdot 9)$
 $= 46.26 \text{ N./mm}^2$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:
 $= Fzp / (\pi \cdot Dz \cdot ez)$
 $= 6631 / (\pi \cdot 4515 \cdot 11)$
 $= 0.42 \text{ N./mm}^2$

Stress [σ_{3qm}]:
 $= Fzq / (\pi \cdot Dz \cdot ez)$
 $= -182971 / (\pi \cdot 4515 \cdot 11)$
 $= -11.50 \text{ N./mm}^2$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:
 $= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$
 $= 1 - 0.5(4517 + 9 - 4515 + 11) / (767 + 9)$
 $= 0.99$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.99) = 9.66 \text{ deg}$$

$$\begin{aligned} \text{Dimension [a]:} \\ &= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))} \\ &= 0.5 \cdot \sqrt{(9^2 + 11^2 + 2 \cdot 9 \cdot 11 \cdot 0.99)} \\ &= 9.96 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Bending moment at verification points p and q [Mp \& Mq]:} \\ M_p &= a \cdot F_{zp} = 9.965 \cdot 6631.3 = 66.08 \text{ Kg-m.} \\ M_q &= a \cdot F_{zq} = 9.965 \cdot -182971.1 = -1823.32 \text{ Kg-m.} \end{aligned}$$

$$\begin{aligned} \text{Correction factor [C]:} \\ &= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{9}{11} \right)^2 \\ &= 0.592 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

$$\begin{aligned} \text{Stress } [\sigma_{1pb(o)} \& \sigma_{2pb(o)}]: \\ &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ &= 0.59 \cdot 6 \cdot 66.1 / (\pi \cdot 4517 \cdot 9^2) \\ &= 2.00 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qb(o)} \& \sigma_{2qb(o)}]: \\ &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ &= 0.59 \cdot 6 \cdot -1823 / (\pi \cdot 4517 \cdot 9^2) \\ &= -55.24 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3pb(o)}]: \\ &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ &= 0.59 \cdot 6 \cdot 66.1 / (\pi \cdot 4515 \cdot 11^2) \\ &= 1.34 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qb(o)}]: \\ &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ &= 0.59 \cdot 6 \cdot -1823 / (\pi \cdot 4515 \cdot 11^2) \\ &= -37.00 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Geometry variables, 16.12-32 to 36:} \\ eb/D_b &= 0.35/178 = 0.002 \\ y &= 125 \cdot eb/D_b = 125 \cdot 9/4517 = 0.25 \end{aligned}$$

As this an elliptical head, it will be treated as a Korbbogen type head

$$\begin{aligned} \text{Geometric value [\alpha]:} \\ &= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994 \\ &= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.25 + 3.994 = 4.52 \end{aligned}$$

$$\begin{aligned} \text{Bending stress in sections 1-1 and 2-2 Equation 16.12-31 } [\sigma_{1b(p)} \& \sigma_{2b(p)}]: \\ &= (P + Ph) \cdot D_b \left(\left(\frac{\gamma}{\gamma_A} \right) \cdot \alpha - 1 \right) / (4 \cdot eb) \\ &= (3.5 + 0) \cdot 4517 \left(\left(\frac{9.66}{40} \right) \cdot 4.52 - 1 \right) / (4 \cdot 9) \\ &= 4.05 \text{ N./mm}^2 \end{aligned}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

$$\begin{aligned} \text{Stress } [\sigma_{1pitot}]: \\ &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 46.8 - 2 + 4.05 \\ &= 48.82 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1potot}]: \\ &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 46.8 + 2 - 4.05 \\ &= 44.72 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qitot}]: \\ &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \end{aligned}$$

$$= 32.2 - -55.2 + 4.05$$

$$= 91.50 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 32.2 + -55.2 - 4.05$$

$$= -27.08 \text{ N./mm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 46.3 + 2 + 4.05$$

$$= 52.31 \text{ N./mm}^2$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 46.3 - 2 - 4.05$$

$$= 40.21 \text{ N./mm}^2$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 46.3 + -55.2 + 4.05$$

$$= -4.93 \text{ N./mm}^2$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 46.3 - -55.2 - 4.05$$

$$= 97.45 \text{ N./mm}^2$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 0.42 - 1.34$$

$$= -0.92 \text{ N./mm}^2$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 0.42 + 1.34$$

$$= 1.76 \text{ N./mm}^2$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -11.5 - -37$$

$$= 25.49 \text{ N./mm}^2$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -11.5 + -37$$

$$= -48.50 \text{ N./mm}^2$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (46.8 / 138)^2$$

$$= 403.43 \text{ N./mm}^2$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (32.2 / 138)^2$$

$$= 408.99 \text{ N./mm}^2$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (46.3 / 138)^2$$

$$= 403.66 \text{ N./mm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (46.3 / 138)^2$$

$$= 403.66 \text{ N./mm}^2$$



For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138 (3 - (1/1.5) \cdot (0.42/138)^2)$$

$$= 414.00 \text{ N./mm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138 (3 - (1/1.5) \cdot (-11.5/138)^2)$$

$$= 413.36 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	48.82	403.43	Pass	46
Stot1po	44.72	403.43	Pass	47
Stot1qi	91.50	408.99	Pass	48
Stot1qo	-27.08	408.99	Pass	49
Stot2pi	52.31	403.66	Pass	50
Stot2po	40.21	403.66	Pass	51
Stot2qi	-4.93	403.66	Pass	52
Stot2qo	97.45	403.66	Pass	53
Stot3pi	-0.92	414.00	Pass	54
Stot3po	1.76	414.00	Pass	55
Stot3qi	25.49	413.36	Pass	56
Stot3qo	-48.50	413.36	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.
Lining is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	9602.736 mm.
Center of Gravity of the Liquid	4828.148 mm.
Center of Gravity of the Lining	7250.000 mm.
Center of Gravity of the Stiffening Rings	7150.000 mm.
Center of Gravity of the Nozzles	7602.930 mm.
Center of Gravity of the Added Weights (Operating)	10250.339 mm.
Center of Gravity of the Added Weights (Empty)	10250.339 mm.
Center of Gravity of Bare Shell New and Cold	4995.567 mm.
Center of Gravity of Bare Shell Corroded	4961.255 mm.
Vessel CG in the Operating Condition	7336.896 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	7312.822 mm.
Vessel CG in the Test Condition	7268.361 mm.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	4526.0000	mm.
Skirt Thickness	STHK	12.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.0000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material		SA-36	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	32.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	N./mm ²
Basing Yield Stress		262.00	N./mm ²
Inside Diameter of Basing	DI, D4	4282.0000	mm.
Outside Diameter of Basing	DOU, D5	4802.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm.
Bolt Corrosion Allowance	BCA	0.0000	mm.
Root Area of a Single Bolt	Area	11.5300	cm ²
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	138.00	N./mm ²
Number of Bolts	RN, nb	20	
Diameter of Bolt Circle	DC, DBC	4662.0000	mm.
Bolt Allowable Shear Stress		82.800	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	41.0	N./mm ²
Allowable Comp. Strength of Concrete	FC	22.5	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm.
Width of Gussets at Top Plate	TWDT	130.0000	mm.
Width of Gussets at Base Plate	BWDT	130.0000	mm.
Gusset Plate Elastic Modulus	E	201734288.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm ²
Height of Chair	h1	350.0000	mm.
Distance between Gussets	RG, b6	96.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	44.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	140.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm.
External Corrosion Allowance	CA	1.6000	mm.
Dead Weight of Vessel	DW	65760.5	Kgf
Operating Weight of Vessel	ROW	96561.3	Kgf
Earthquake Moment on Basing	EQMOM	91311.4	Kg-m.
Wind Moment on Basing	WIMOM	22970.7	Kg-m.
User Moment at the Base [Earthquake Case]		112645.8	Kg-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		112645.8	Kg-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 203957/4662) - 94326) / 20$$

$$= 4033.3 \text{ Kg}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 135616/4662) - 65761) / 20$$

$$= 2529.8 \text{ Kg}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/4662) - 96561) / 20$$

$$= -4828.1 \text{ Kg}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in one Bolt Abss : 11.5300 cm²
 Area Available in all the Bolts Abss * RN : 230.6000 cm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.300	0.451	1.511	2.441	0.438	0.781	3169.0	97494.9	228.2
4	0.526	0.488	2.058	1.941	0.387	0.785	9265.1	103591.0	839.1
6	0.469	0.460	1.930	2.069	0.400	0.785	7686.1	102012.1	652.9
8	0.455	0.458	1.897	2.101	0.403	0.785	7298.2	101624.1	610.6
10	0.459	0.459	1.905	2.093	0.402	0.785	7394.9	101720.9	621.1
12	0.460	0.460	1.907	2.091	0.402	0.785	7419.1	101745.1	623.7
13	0.460	0.460	1.907	2.091	0.402	0.785	7415.1	101741.1	623.2

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 7415 / (1.57 \cdot 4662 \cdot 2.09)$$

$$= 9.47 \text{ N./mm}^2, \text{ Should be } \leq 138 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot Bolt \text{ Area} / (\pi \cdot Dc)$$

$$= 20 \cdot 11.5 / (\pi \cdot 4662)$$

$$= 1.57 \text{ mm.}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 0.82[(2 \cdot 0.46 \cdot 4662 + 260) / (2 \cdot 0.46 \cdot 4662)]$$

$$= 0.87 \text{ N./mm}^2, \text{ Should be } \leq 22.5 \text{ N./mm}^2$$

Values for table 10.3, l = 138, b = 366, l/b = 0.37689179

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(159, 674) = 674 \text{ Kg}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 674 / 173)} + 1.6$$

$$= 16.7 \text{ mm.}$$

Nomenclature:

a = (Dc-Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 140, b = 96, b/l = 0.6857143
 As b/l (0.69) is less than 1, inverting b/l = 1.46 0.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 16225 / (4 \cdot \pi) [1.3(\ln((2 \cdot 140 \cdot \sin(\pi \cdot 68/140))) / (\pi \cdot 39.7))) + 1] - [(0.7 - 0.076) \cdot 16225 / (4 \cdot \pi)]$$

= 1841.0519 Kgf

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{\frac{6 \cdot \text{Abs}(\text{Mo})}{F_b}} + C_a$$

$$= \sqrt{\frac{6 \cdot \text{Abs}(1841)}{175}} + 1.6$$

$$= 26.5042 \text{ mm.}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot l_{\text{avg}}) + C_a$$

$$= 16225 / (124 \cdot 130) + 1.6$$

$$= 11.5 \text{ (not less than } 9.52 + 1.6 \text{) mm.}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 4662 / 20 = 732 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot S_{ktope}))^{2/3} \cdot r^{1/3} + C_a$$

$$= 1.76 (16225 \cdot 68 / (732 \cdot 306 \cdot 171))^{2/3} \cdot 22631^{1/3} + C_a$$

$$= 9.95 + 1 = 10.9 \text{ mm.}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 19505 / (11.5 \cdot 20)$$

$$= 8.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	16.7493	mm.
Actual Basing Thickness as entered by user	32.0000	mm.
Required Thickness of Chair Cap	26.5042	mm.
Actual Top Ring Thickness as entered by user	44.0000	mm.
Required Gusset thickness, + CA	11.4618	mm.
Actual Gusset Thickness as entered by user	16.0000	mm.
Required Thickness of Skirt for Local Stress	10.9453	mm.
Given Thickness of Skirt	12.0000	mm.
Required Gusset Height to meet local stress	254.1526	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - C_A)$$

$$= 17.5 \cdot (12 - 1)$$

$$= 19.7 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 19.7 / [(0.4 \cdot 225) \cdot 2 \cdot 0.707]$$

$$= 1.51 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cm}w_{th} + 2 \cdot (H_g + T_{ta}))$$

$$= 16225 / (178 + 2 \cdot (274 + 44))$$

$$= 19.9 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cm}w_{th} \cdot (H_g + T_{ta}) + 0.6667 \cdot (H_g + T_{ta})^2)$$

$$= 16225 \cdot 68 / (178 \cdot (318) + 0.6667 \cdot (318)^2)$$

$$= 8.9 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{W_v^2 + W_h^2}$$

$$= \sqrt{19.9^2 + 8.9^2}$$

$$= 21.8 \text{ Kg/mm.}$$



Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 21.8 / [(0.4 \cdot 225) \cdot 2 \cdot 0.707] \\ &= 1.68 \text{ mm.} \end{aligned}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$\begin{aligned} &= \text{Bolt Load} / (2 \cdot \text{TopWth}) \\ &= 16225 / (2 \cdot 130) \\ &= 62.4 \text{ Kg/mm.} \end{aligned}$$

Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 16225 \cdot 68 / (2 \cdot 318 \cdot 130) \\ &= 13.3 \text{ Kg/mm.} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{W_v^2 + W_h^2} \\ &= \sqrt{62.4^2 + 13.3^2} \\ &= 63.8 \text{ Kg/mm.} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 63.8 / [(0.4 \cdot 225) \cdot 2 \cdot 0.707] \\ &= 4.91 \text{ mm.} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	4.7625	mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	4.9089	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N3-4"

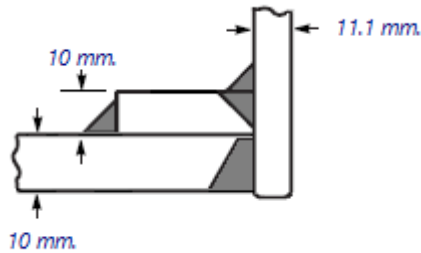
From: 40

Pressure for Reinforcement Calculations	P	3.679	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	4506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	220.0000	mm.
Thickness of Pad	te	10.0000	mm.
Weld leg size between Pad and Shell	Wp	10.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm.
Reinforcing Pad Width		52.8500	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2500.0	mm.
Centerline Distance to Tangent		1500.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N3-4"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.68 \cdot 4508 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.68)$
 $= 5.4071$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.68 \cdot 57.1 / (118 \cdot 1 + 0.4 \cdot 3.68)$
 $= 0.1780$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3853 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	188.0992 mm.
Parallel to Vessel Wall, opening length	d	94.0496 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

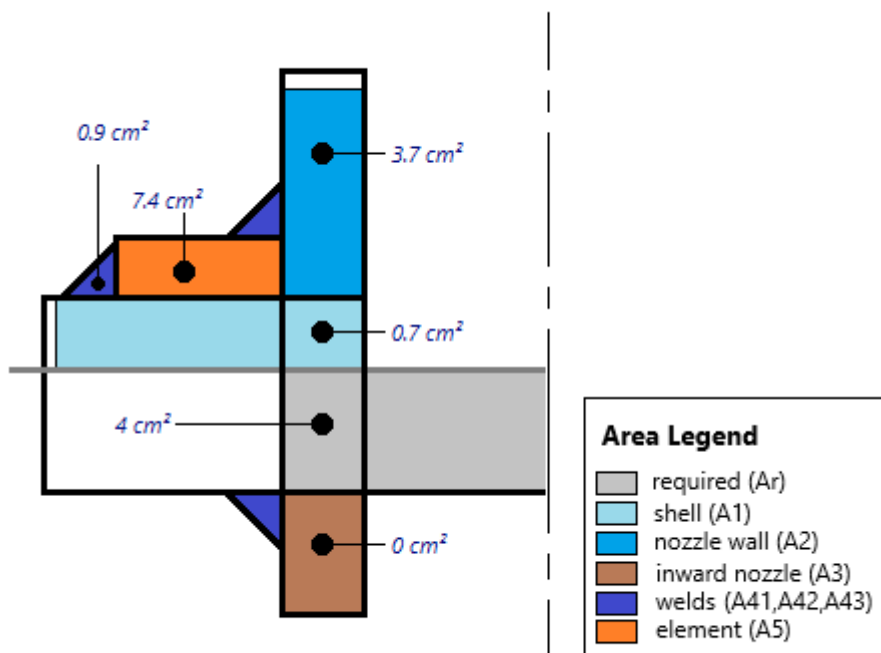


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5.244	4.018	NA
Area in Shell	A1	3.274	0.650	NA
Area in Nozzle Wall	A2	3.827	3.748	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	7.380	7.380	NA
TOTAL AREA AVAILABLE	Atot	15.336	12.633	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(94 \cdot 8.29 \cdot 1 + 2 \cdot 10.1 \cdot 8.29 \cdot 1(1 - 0.86))$
 $= 4.018 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 94(1 \cdot 9 - 1 \cdot 8.29) - 2 \cdot 10.1$
 $(1 \cdot 9 - 1 \cdot 8.29) \cdot (1 - 0.86)$
 $= 0.650 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 22.5) \cdot (10.1 - 0.39) \cdot 0.86$
 $= 3.748 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1$

= 0.855 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
 = (188 - 114) • 10 • 1
 = 7.380 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 1.3853 mm.
 Wall Thickness per UG16(b), tr16b = 2.5000 mm.
 Wall Thickness, shell/head, internal pressure trb1 = 7.0069 mm.
 Wall Thickness tb1 = max(trb1, tr16b) = 7.0069 mm.
 Wall Thickness, shell/head, external pressure trb2 = 1.8441 mm.
 Wall Thickness tb2 = max(trb2, tr16b) = 2.5000 mm.
 Wall Thickness per table UG-45 tb3 = 6.2578 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[6.26, max(7.01, 2.5)]
 = 6.2578 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(1.39, 6.26)
 = 6.2578 mm.

Available Nozzle Neck Thickness = 0.875 * 11.125 = 9.735 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	28.0,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	267.0 N./mm ²	Passed
Occasional	:	0.8,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	18.9,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.73, tr = 0.18, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.02, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.73, tr = 0.18, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.02, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.73, tr = 0.18, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.02, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 10, tr = 5.41, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.6, Temp. Reduction = 22 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 9.73, tr = 0.18, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.02, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.68/51.00 = 0.072

Weld Size Calculations, Description: N3-4"

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 10.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	5.0000 = 0.5*TminPad	7.0700 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (4.02 - 0.65 + 2 \cdot 10.1 \cdot 0.86 \cdot (1 \cdot 9 - 8.29)) \cdot 138)$
 $= 4913.12 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (3.75 + 7.38 + 0.86 - 0 \cdot 0.86) \cdot 138$
 $= 16861.61 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (3.75 + 0 + 0.86 + (1.56)) \cdot 138$
 $= 8669.81 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (3.75 + 0 + 0.86 + 7.38 + (1.56)) \cdot 138$
 $= 19054.52 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 118$
 $= 10585. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 220 \cdot 10 \cdot 0.49 \cdot 138$
 $= 23828. \text{ Kg}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 52.1) \cdot (11.1 - 1) \cdot 0.7 \cdot 118$$

$$= 13955. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 114 \cdot 10 \cdot 0.74 \cdot 138$$

$$= 18696. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 114 \cdot (10 - 1) \cdot 0.74 \cdot 138$$

$$= 16826. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (23828 + 13955) = 37783 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (10585 + 18696 + 16826 + 0) = 46107 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (23828 + 16826 + 0) = 40654 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 37782	Kgf, must exceed W = 4913 Kg	or W1 = 16861 Kg
Path 2-2 = 46107	Kgf, must exceed W = 4913 Kg	or W2 = 8669 Kg
Path 3-3 = 40654	Kgf, must exceed W = 4913 Kg	or W3 = 19054 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 0.4019 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 260.4021 mm.

Input Echo, WRC107/537 Item 1, Description: N3-4" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm.
Vessel Diameter	Dv	8110.800	mm.
Vessel Thickness	Tv	10.000	mm.
Design Temperature	T1	110.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	1.0000 mm.
Nozzle Diameter	Dn	114.300 mm.
Nozzle Thickness	Tn	11.125 mm.
Nozzle Material		SA-106 B



Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

Thickness of Reinforcing Pad	Tpad	10.000	mm.
Diameter of Reinforcing Pad	Dpad	220.000	mm.

Design Internal Pressure	Dp	3.679	bars
Include Pressure Thrust		No	

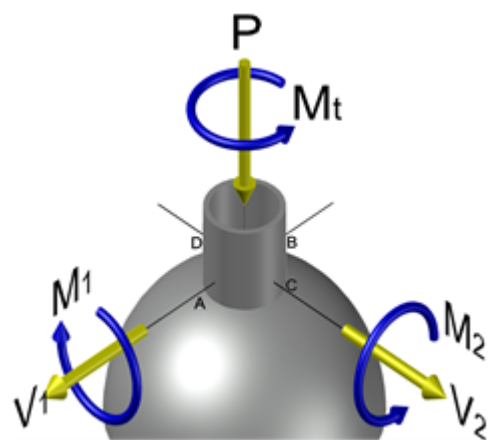
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	764.5	Kgf
Longitudinal Shear	(SUS)	(V1) V1	509.7	Kgf
Circumferential Shear	(SUS)	(Vc) V2	509.7	Kgf
Circumferential Moment	(SUS)	(Mc) M1	142.7	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	142.7	Kg-m.
Torsional Moment	(SUS)	Mt	203.9	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 114.3 + 2 \cdot 1.65 \cdot \sqrt{4060.9 (10.0 - 1.0)} \\
 &= 745.179 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	764.5	Kgf
Circumferential Shear	(VC)	V2	509.7	Kgf
Longitudinal Shear	(VL)	V1	509.7	Kgf
Circumferential Moment	(MC)	M1	142.7	Kg-m.
Longitudinal Moment	(ML)	M2	142.7	Kg-m.
Torsional Moment		MT	203.9	Kg-m.

Dimensionless Parameters: U = 0.21 TAU = 5.14 RHO = 1.88

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
----------------------	--------	-------	----------



N(x) * T / P	SP 2	0.13315	(A,B,C,D)
M(x) / P	SP 2	0.11559	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 2	0.39490	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 2	0.40537	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 2	0.39490	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 2	0.40537	(A,B,C,D)
N(y) * T / P	SP 2	0.22574	(A,B,C,D)
M(y) / P	SP 2	0.10240	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 2	0.07734	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 2	0.52988	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 2	0.07734	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 2	0.52988	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-2.8	-2.8	-2.8	-2.8	-2.8	-2.8	-2.8	-2.8
Rad. Bend. P		-14.4	14.4	-14.4	14.4	-14.4	14.4	-14.4	14.4
Rad. Memb. MC		0.0	0.0	0.0	0.0	-5.5	-5.5	5.5	5.5
Rad. Bend. MC		0.0	0.0	0.0	0.0	-33.9	33.9	33.9	-33.9
Rad. Memb. ML		-5.5	-5.5	5.5	5.5	0.0	0.0	0.0	0.0
Rad. Bend. ML		-33.9	33.9	33.9	-33.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-56.6	40.1	22.3	-16.8	-56.6	40.1	22.3	-16.8
Tang. Memb. P		-4.7	-4.7	-4.7	-4.7	-4.7	-4.7	-4.7	-4.7
Tang. Bend. P		-12.8	12.8	-12.8	12.8	-12.8	12.8	-12.8	12.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Tang. Bend. MC		0.0	0.0	0.0	0.0	-44.3	44.3	44.3	-44.3
Tang. Memb. ML		-1.1	-1.1	1.1	1.1	0.0	0.0	0.0	0.0
Tang. Bend. ML		-44.3	44.3	44.3	-44.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-62.9	51.3	28.0	-35.2	-62.9	51.3	28.0	-35.2
Shear VC		1.5	1.5	-1.5	-1.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Shear MT		5.1	5.1	5.1	5.1	5.1	5.1	5.1	5.1
Tot. Shear		6.6	6.6	3.7	3.7	3.7	3.7	6.6	6.6
Str. Int.		67.0	54.4	29.8	35.9	64.6	52.4	32.3	37.3

Unitless Prm: U = 0.58 TAU = 0.00 (10.36) RHO = 0.00 (0.89)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.11962	(A,B,C,D)
M(x) / P	SR 2	0.07087	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.13214	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.19893	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.13214	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.19893	(A,B,C,D)
N(y) * T / P	SR 2	0.03585	(A,B,C,D)
M(y) / P	SR 2	0.02162	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.03998	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.05931	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.03998	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.05931	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-11.1	-11.1	-11.1	-11.1	-11.1	-11.1	-11.1	-11.1
Rad. Bend. P		-39.4	39.4	-39.4	39.4	-39.4	39.4	-39.4	39.4
Rad. Memb. MC		0.0	0.0	0.0	0.0	-11.9	-11.9	11.9	11.9
Rad. Bend. MC		0.0	0.0	0.0	0.0	-107.9	107.9	107.9	-107.9
Rad. Memb. ML		-11.9	-11.9	11.9	11.9	0.0	0.0	0.0	0.0
Rad. Bend. ML		-107.9	107.9	107.9	-107.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-170.3	124.2	69.4	-67.6	-170.3	124.2	69.4	-67.6
Tang. Memb. P		-3.3	-3.3	-3.3	-3.3	-3.3	-3.3	-3.3	-3.3
Tang. Bend. P		-12.0	12.0	-12.0	12.0	-12.0	12.0	-12.0	12.0
Tang. Memb. MC		0.0	0.0	0.0	0.0	-3.6	-3.6	3.6	3.6
Tang. Bend. MC		0.0	0.0	0.0	0.0	-32.2	32.2	32.2	-32.2
Tang. Memb. ML		-3.6	-3.6	3.6	3.6	0.0	0.0	0.0	0.0
Tang. Bend. ML		-32.2	32.2	32.2	-32.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-51.1	37.2	20.4	-19.9	-51.1	37.2	20.4	-19.9
Shear VC		1.6	1.6	-1.6	-1.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Shear MT		2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
Tot. Shear		4.5	4.5	1.3	1.3	1.3	1.3	4.5	4.5
Str. Int.		170.4	124.5	69.4	67.7	170.3	124.2	69.8	68.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		39.3	39.3	39.3	39.3	39.3	39.3	39.3	39.3
Rad. Pl (SUS)		-8.3	-8.3	2.7	2.7	-8.3	-8.3	2.7	2.7
Rad. Q (SUS)		-48.3	48.3	19.5	-19.5	-48.3	48.3	19.5	-19.5
Long. Pm (SUS)		39.3	39.3	39.3	39.3	39.3	39.3	39.3	39.3
Long. Pl (SUS)		-5.8	-5.8	-3.6	-3.6	-5.8	-5.8	-3.6	-3.6
Long. Q (SUS)		-57.1	57.1	31.6	-31.6	-57.1	57.1	31.6	-31.6
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.5	1.5	-1.5	-1.5	-1.5	-1.5	1.5	1.5
Shear Q (SUS)		5.1	5.1	5.1	5.1	5.1	5.1	5.1	5.1
Pm (SUS)		39.3	39.3	39.3	39.3	39.3	39.3	39.3	39.3
Pm+Pl (SUS)		34.2	34.2	42.3	42.3	34.2	34.2	42.3	42.3
Pm+Pl+Q (Total)		27.8	93.6	69.0	23.2	25.3	91.7	71.6	24.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.27	138.00	Passed
Pm+Pl (SUS)	42.34	207.00	Passed
Pm+Pl+Q (TOTAL)	93.65	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	82.9	82.9	82.9	82.9	82.9	82.9	82.9	82.9
Rad.	Pl (SUS)	-23.0	-23.0	0.9	0.9	-23.0	-23.0	0.9	0.9
Rad.	Q (SUS)	-147.2	147.2	68.5	-68.5	-147.2	147.2	68.5	-68.5
Long.	Pm (SUS)	82.9	82.9	82.9	82.9	82.9	82.9	82.9	82.9
Long.	Pl (SUS)	-6.9	-6.9	0.3	0.3	-6.9	-6.9	0.3	0.3
Long.	Q (SUS)	-44.2	44.2	20.2	-20.2	-44.2	44.2	20.2	-20.2
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.6	1.6	-1.6	-1.6	-1.6	-1.6	1.6	1.6
Shear	Q (SUS)	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
Pm (SUS)		82.9	82.9	82.9	82.9	82.9	82.9	82.9	82.9
Pm+Pl (SUS)		76.1	76.1	85.1	85.1	76.1	76.1	85.1	85.1
Pm+Pl+Q (Total)		119.5	207.4	152.3	63.1	119.2	207.2	152.7	63.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	82.90	138.00	Passed
Pm+Pl (SUS)	85.12	207.00	Passed
Pm+Pl+Q (TOTAL)	207.37	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N1-42"

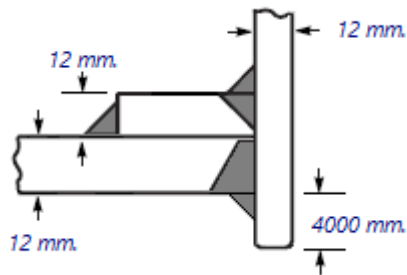
From: 50

Pressure for Reinforcement Calculations	P	3.507	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4506.00	mm.
Design Length of Section	L	3200.0000	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2050.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		42.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	4000.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1450.8000	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		180.0001	mm.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N1-42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	42.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.51 \cdot 2254 / (138 \cdot 1 - 0.6 \cdot 3.51)$
 $= 5.7369$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.51 \cdot 534 / (138 \cdot 1 - 0.6 \cdot 3.51)$
 $= 1.3601$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.6497 mm.

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	2137.5999 mm.
Parallel to Vessel Wall, opening length	d	1068.7999 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

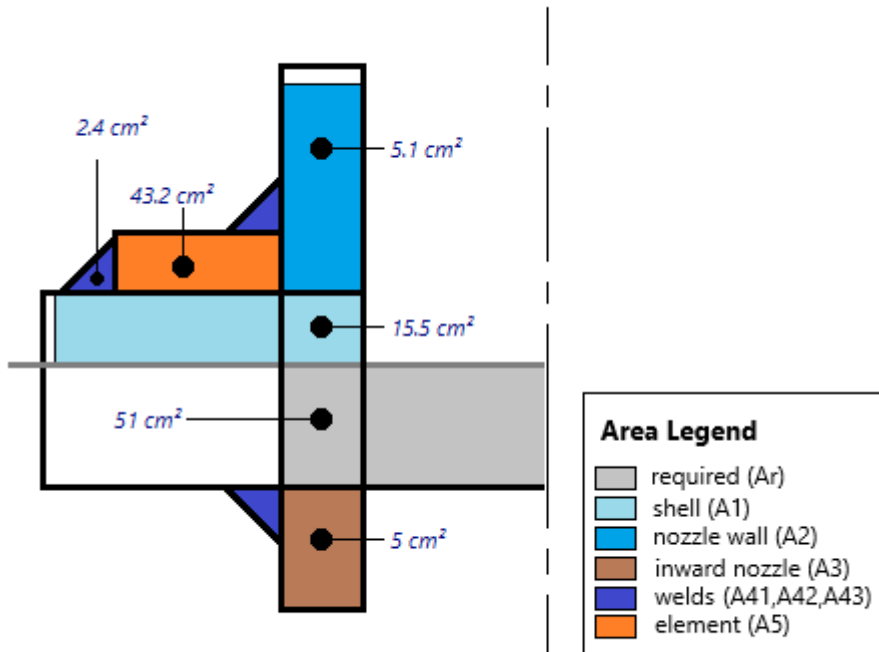


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	61.315	51.048	NA
Area in Shell	A1	56.253	15.472	NA
Area in Nozzle Wall	A2	5.302	5.143	NA
Area in Inward Nozzle	A3	5.000	5.000	NA
Area in Welds	A41+A42+A43	2.440	2.440	NA
Area in Element	A5	43.200	43.200	NA
TOTAL AREA AVAILABLE	Atot	112.194	71.254	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1282.4171 12.0000 mm.
 Based on given Pad Diameter: 1450.8000 6.3872 mm.
 Based on Shell or Nozzle Thickness: 1282.4171 12.0000 mm.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (1069 \cdot 9.55 \cdot 1 + 2 \cdot 11 \cdot 9.55 \cdot 1 (1 - 1))$
 $= 51.048 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 1069 (1 \cdot 11 - 1 \cdot 9.55) - 2 \cdot 11$
 $(1 \cdot 11 - 1 \cdot 9.55) \cdot (1 - 1)$
 $= 15.472 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 27.5) \cdot (11 - 1.65) \cdot 1$

$$= 5.143 \text{ cm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot t_i \cdot \min(h, Tl, 2.5 \cdot t_i) \cdot fr2$$

$$= 2 \cdot 10 \cdot (25) \cdot 1$$

$$= 5.000 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 12^2 \cdot 1$$

$$= 2.440 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (1451 - 1091) \cdot 12 \cdot 1$$

$$= 43.200 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.6497 mm.
Wall Thickness per UG16(b),	tr16b = 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 6.7369 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 6.7369 mm.
Wall Thickness, shell/head, external pressure	trb2 = 1.8448 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm.
Wall Thickness per table UG-45	tb3 = 9.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.33, \max(6.74, 2.5)]$$

$$= 6.7369 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.65, 6.74)$$

$$= 6.7369 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	32.4,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	312.6 N./mm ²	Passed
Occasional	:	8.4,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	15.2,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

$$\text{Govrn. thk, } tg = 12, tr = 1.36, c = 1 \text{ mm., } E* = 1$$

$$\text{Thickness Ratio} = tr \cdot E*/(tg - c) = 0.12, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

$$\text{Govrn. thk, } tg = 12, tr = 1.36, c = 1 \text{ mm., } E* = 1$$

$$\text{Thickness Ratio} = tr \cdot E*/(tg - c) = 0.12, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

$$\text{Govrn. thk, } tg = 12, tr = 1.36, c = 1 \text{ mm., } E* = 1$$



Thickness Ratio = $tr * E^* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $tg = 12$, $tr = 5.74$, $c = 1$ mm., $E^* = 1$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $tg = 12$, $tr = 5.74$, $c = 1$ mm., $E^* = 1$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.51/19.60 = 0.179$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1603.2000 mm.
 Parallel to Vessel Wall, opening length d 801.6000 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 27.5000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [External Pressure]	AR	34.034	34.034
Area Available in Shell	A1	7.736	7.736
Area Available in Nozzle Wall	A2	5.143	5.143
Area Available in Inward Nozzle	A3	5.000	5.000
Area Available in Welds	A4	1.000	2.440
Area Available in Pad	A5	0.000	43.200
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	18.878	63.518

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

Warning: Ins. Proj. Specified -- This calculation must comply with U-2(g).

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

Warning: The flange could not be considered in the moment of inertia calculations.

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

	Area	Distance	Inertia	d	I+Ad ²
Nozzle	21.8900	99.5000	722.3875	57.115	1436.4563
Shell	19.3600	5.5000	1.9521	36.885	265.3505
Pad	21.1200	17.0000	2.5344	25.385	138.6351
Sum	62.3700				1840.44

Membrane stress per Appendix 1-7 (1 or 2) Sm 121.54 N./mm²
 Allowable Membrane Stress S 138.00 N./mm²



Bending stress at the inside of the Nozzle	Sb	56.43	N./mm ²
Combined Membrane plus Bending Stress	Sm + Sb	177.97	N./mm ²
Allowable Membrane + Bending Stress	1.5 * S	207.00	N./mm ²
M.A.W.P for the Appendix 1-7 Stress calc		3.98	bars

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle	=	11.007	cm ²
Area from Shell	=	17.342	cm ²
Area from Pad	=	18.918	cm ²
Total Area, As	=	47.267	cm ²

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P \left[R \left(R_n + t_n + \sqrt{r_m \cdot t} \right) + R_n \left(t + t_e + \sqrt{R_n m \cdot t_n} \right) \right] / A_s$$

$$= 3.51 \left[2254 \left(534 + 11 + 158 \right) + 534 \left(100 \right) \right] / 47.3$$

$$= 122 \text{ N./mm}^2$$

The area As was computed per figure 1-7-1 for the Sm calculation.

Moment [M]:

$$= \left(R_n^3 / 6 + R \cdot R_n \cdot e \right) P$$

$$= \left(534^3 / 6 + 2254 \cdot 534 \cdot 36.9 \right) / 3.51$$

$$= 2499 \text{ Kg-m.}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= \left(M \left(e + t/2 \right) \right) / I$$

$$= \left(2499 \left(36.9 + 11/2 \right) \right) / 1840$$

$$= 56.428 \text{ N./mm}^2$$

Weld Size Calculations, Description: N1-42"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.0000	mm.
Intermediate Calc. for pad/shell Welds	TminPad	11.0000	mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	5.5000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max \left(0, \left(A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r) \right) \cdot S_v \right)$$

$$= \max \left(0, \left(51 - 15.5 + 2 \cdot 11 \cdot 1 \cdot (1 \cdot 11 - 9.55) \right) \cdot 138 \right)$$

$$= 50511.07 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= \left(A_2 + A_5 + A_4 - (W_i - C_{an} / .707)^2 \cdot f_{r2} \right) \cdot S_v$$

$$= \left(5.14 + 43.2 + 2.44 - 0 \cdot 1 \right) \cdot 138$$

$$= 71460.48 \text{ Kg}$$

Weld Load [W2]:

$$= \left(A_2 + A_3 + A_4 + \left(2 \cdot t_n \cdot t \cdot f_{r1} \right) \right) \cdot S_v$$

$$= \left(5.14 + 5 + 1 + \left(2.42 \right) \right) \cdot 138$$

$$= 19085.14 \text{ Kg}$$

Weld Load [W3]:

$$= \left(A_2 + A_3 + A_4 + A_5 + \left(2 \cdot t_n \cdot t \cdot f_{r1} \right) \right) \cdot S$$

$$= \left(5.14 + 5 + 2.44 + 43.2 + \left(2.42 \right) \right) \cdot 138$$

$$= 81901.78 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= \left(\pi / 2 \right) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= \left(3.14 / 2.0 \right) \cdot 1091 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 118144. \text{ Kg}$$



Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 1451 \cdot 12 \cdot 0.49 \cdot 138$
 $= 188562. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 540) \cdot (12 - 1) \cdot 0.7 \cdot 138$
 $= 183783. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 1091 \cdot 12 \cdot 0.74 \cdot 138$
 $= 214106. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 1091 \cdot (12 - 1) \cdot 0.74 \cdot 138$
 $= 196264. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (188562 + 183783) = 372345 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (118144 + 214106 + 196264 + 0) = 528513 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (188562 + 196264 + 0) = 384826 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 372345 Kg, must exceed W = 50511 Kg or W1 = 71460 Kg
 Path 2-2 = 528513 Kg, must exceed W = 50511 Kg or W2 = 19085 Kg
 Path 3-3 = 384826 Kg, must exceed W = 50511 Kg or W3 = 81901 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 67.0110 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 4362.0000 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 1.112 %

Warning:
 Review Appendix 1-7 (b)(3) for notes regarding externally applied loads
 to large nozzles and U-2(g).

Input Echo, WRC107/537 Item 1, Description: N1-42" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	1.0000 mm.
Vessel Diameter	Dv	4506.000 mm.
Vessel Thickness	Tv	12.000 mm.
Design Temperature	T1	110.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²



Vessel Hot S.I. Allowable Smh 138.00 N./mm²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

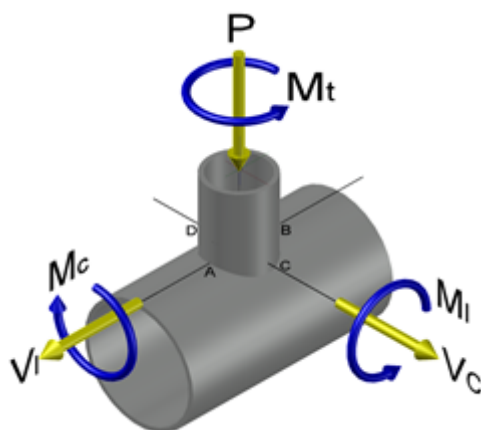
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.0000	mm.
Nozzle Diameter	Dn	1066.800	mm.
Nozzle Thickness	Tn	12.000	mm.
Nozzle Material		SA-516 70	
Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	12.000	mm.
Diameter of Reinforcing Pad	Dpad	1450.800	mm.
Design Internal Pressure	Dp	3.507	bars
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	8030.8	Kgf
Longitudinal Shear (SUS)	Vl	5353.9	Kgf
Circumferential Shear (SUS)	Vc	5353.9	Kgf
Circumferential Moment (SUS)	Mc	15734.3	Kg-m.
Longitudinal Moment (SUS)	Ml	15734.3	Kg-m.
Torsional Moment (SUS)	Mt	22477.6	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979
Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1090.8 + 2 \cdot 1.65 \cdot \sqrt{(2259.5 (12.0 - 1.0))}$$

$$= 1611.055 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	8030.8	Kgf
-------------	---	--------	-----



Circumferential Shear	VC	5353.9	Kgf
Longitudinal Shear	VL	5353.9	Kgf
Circumferential Moment	MC	15734.3	Kg-m.
Longitudinal Moment	ML	15734.3	Kg-m.
Torsional Moment	MT	22477.6	Kg-m.

Dimensionless Parameters used : Gamma = 98.50

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.211	4C	10.531	(A,B)
N(PHI) / (P/Rm)	0.211	3C	4.715	(C,D)
M(PHI) / (P)	0.211	2C1	0.016	(A,B)
M(PHI) / (P)	0.211	1C !	0.066	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.211	3A	3.129	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.211	1A	0.056	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.211	3B	6.975	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.211	1B	0.011	(A,B,C,D)
N(x) / (P/Rm)	0.211	3C	4.715	(A,B)
N(x) / (P/Rm)	0.211	4C	10.531	(C,D)
M(x) / (P)	0.211	1C1	0.039	(A,B)
M(x) / (P)	0.211	2C !	0.036	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.211	4A	8.840	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.211	2A	0.024	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.211	4B	3.332	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.211	2B	0.014	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-15.9	-15.9	-15.9	-15.9	-7.1	-7.1	-7.1	-7.1
Circ. Bend. P		-13.9	13.9	-13.9	13.9	-59.4	59.4	-59.4	59.4
Circ. Memb. MC		0.0	0.0	0.0	0.0	-19.4	-19.4	19.4	19.4
Circ. Bend. MC		0.0	0.0	0.0	0.0	-205.6	205.6	205.6	-205.6
Circ. Memb. ML		-43.3	-43.3	43.3	43.3	0.0	0.0	0.0	0.0
Circ. Bend. ML		-39.5	39.5	39.5	-39.5	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-112.6	-5.8	53.0	1.8	-291.6	238.5	158.5	-133.9
Long. Memb. P		-7.1	-7.1	-7.1	-7.1	-15.9	-15.9	-15.9	-15.9
Long. Bend. P		-34.5	34.5	-34.5	34.5	-32.3	32.3	-32.3	32.3
Long. Memb. MC		0.0	0.0	0.0	0.0	-54.9	-54.9	54.9	54.9
Long. Bend. MC		0.0	0.0	0.0	0.0	-87.9	87.9	87.9	-87.9
Long. Memb. ML		-20.7	-20.7	20.7	20.7	0.0	0.0	0.0	0.0
Long. Bend. ML		-52.7	52.7	52.7	-52.7	0.0	0.0	0.0	0.0
Tot. Long. Str.		-115.0	59.4	31.7	-4.6	-191.0	49.4	94.4	-16.6
Shear VC		1.3	1.3	-1.3	-1.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Shear MT		5.1	5.1	5.1	5.1	5.1	5.1	5.1	5.1
Tot. Shear		6.5	6.5	3.8	3.8	3.8	3.8	6.5	6.5
Str. Int.		120.4	66.5	53.6	9.9	291.7	238.5	159.2	134.3

Dimensionless Parameters used : Gamma = 205.41

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.281	4C	12.533	(A,B)
N(PHI) / (P/Rm)	0.281	3C	3.175	(C,D)
M(PHI) / (P)	0.281	2C1	0.005	(A,B)
M(PHI) / (P)	0.281	1C !	0.068	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.281	3A	3.561	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.281	1A	0.047	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.281	3B	7.005	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.281	1B	0.003	(A,B,C,D)
N(x) / (P/Rm)	0.281	3C	3.175	(A,B)
N(x) / (P/Rm)	0.281	4C	12.533	(C,D)
M(x) / (P)	0.281	1C1	0.014	(A,B)
M(x) / (P)	0.281	2C !	0.037	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.281	4A	15.461	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.281	2A	0.017	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.281	4B	3.177	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.281	2B	0.004	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-39.7	-39.7	-39.7	-39.7	-10.1	-10.1	-10.1	-10.1
Circ. Bend. P		-18.3	18.3	-18.3	18.3	-264.6	264.6	-264.6	264.6
Circ. Memb. MC		0.0	0.0	0.0	0.0	-34.8	-34.8	34.8	34.8
Circ. Bend. MC		0.0	0.0	0.0	0.0	-571.8	571.8	571.8	-571.8
Circ. Memb. ML		-68.5	-68.5	68.5	68.5	0.0	0.0	0.0	0.0
Circ. Bend. ML		-41.3	41.3	41.3	-41.3	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-167.8	-48.6	51.8	5.8	-881.3	791.6	331.9	-282.4
Long. Memb. P		-10.1	-10.1	-10.1	-10.1	-39.7	-39.7	-39.7	-39.7
Long. Bend. P		-56.4	56.4	-56.4	56.4	-144.1	144.1	-144.1	144.1
Long. Memb. MC		0.0	0.0	0.0	0.0	-151.2	-151.2	151.2	151.2
Long. Bend. MC		0.0	0.0	0.0	0.0	-206.8	206.8	206.8	-206.8
Long. Memb. ML		-31.1	-31.1	31.1	31.1	0.0	0.0	0.0	0.0
Long. Bend. ML		-46.3	46.3	46.3	-46.3	0.0	0.0	0.0	0.0
Tot. Long. Str.		-143.9	61.6	11.0	31.1	-541.8	160.0	174.3	48.8
Shear VC		2.1	2.1	-2.1	-2.1	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Shear MT		6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Tot. Shear		8.2	8.2	4.0	4.0	4.0	4.0	8.2	8.2
Str. Int.		170.3	111.4	52.2	31.7	881.4	791.6	332.3	331.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	34.2	34.5	34.2	34.5	34.2	34.5	34.2	34.5
Circ. Pl (SUS)	-59.2	-59.2	27.4	27.4	-26.5	-26.5	12.3	12.3
Circ. Q (SUS)	-53.4	53.4	25.6	-25.6	-265.0	265.0	146.2	-146.2
Long. Pm (SUS)	17.1	17.1	17.1	17.1	17.1	17.1	17.1	17.1
Long. Pl (SUS)	-27.8	-27.8	13.5	13.5	-70.8	-70.8	38.9	38.9
Long. Q (SUS)	-87.2	87.2	18.2	-18.2	-120.2	120.2	55.5	-55.5
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	1.3	1.3	-1.3	-1.3	-1.3	-1.3	1.3	1.3
Shear Q (SUS)	5.1	5.1	5.1	5.1	5.1	5.1	5.1	5.1
Pm (SUS)	34.2	34.5	34.2	34.5	34.2	34.5	34.2	34.5
Pm+Pl (SUS)	25.1	24.8	61.6	62.0	61.4	61.7	56.2	56.2
Pm+Pl+Q (Total)	99.9	77.4	87.5	36.9	257.5	273.1	193.2	100.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	34.54	138.00	Passed
Pm+Pl (SUS)	61.97	207.00	Passed
Pm+Pl+Q (TOTAL)	273.09	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		71.7	72.0	71.7	72.0	71.7	72.0	71.7	72.0
Circ. Pl (SUS)		-108.2	-108.2	28.8	28.8	-44.9	-44.9	24.8	24.8
Circ. Q (SUS)		-59.6	59.6	23.0	-23.0	-836.4	836.4	307.2	-307.2
Long. Pm (SUS)		35.8	35.8	35.8	35.8	35.8	35.8	35.8	35.8
Long. Pl (SUS)		-41.1	-41.1	21.0	21.0	-190.9	-190.9	111.5	111.5
Long. Q (SUS)		-102.7	102.7	-10.1	10.1	-350.9	350.9	62.7	-62.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		2.1	2.1	-2.1	-2.1	-2.1	-2.1	2.1	2.1
Shear Q (SUS)		6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Pm (SUS)		71.7	72.0	71.7	72.0	71.7	72.0	71.7	72.0
Pm+Pl (SUS)		36.7	36.3	100.6	100.9	181.9	182.3	147.4	147.4
Pm+Pl+Q (Total)		112.2	98.3	123.7	79.1	809.7	863.6	404.0	295.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	72.04	138.00	Passed
Pm+Pl (SUS)	182.29	207.00	Passed
Pm+Pl+Q (TOTAL)	863.62	414.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N4-2"

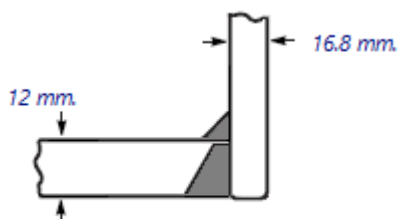
From: 50

Pressure for Reinforcement Calculations	P	3.507	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4506.00	mm.
Design Length of Section	L	3200.0000	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		900.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.51 \cdot 2254 / (138 \cdot 1 - 0.6 \cdot 3.51)$
 $= 5.7369$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.51 / (115 \cdot 1)] - 1))$
 $= 0.0806$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2756 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	106.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	53.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

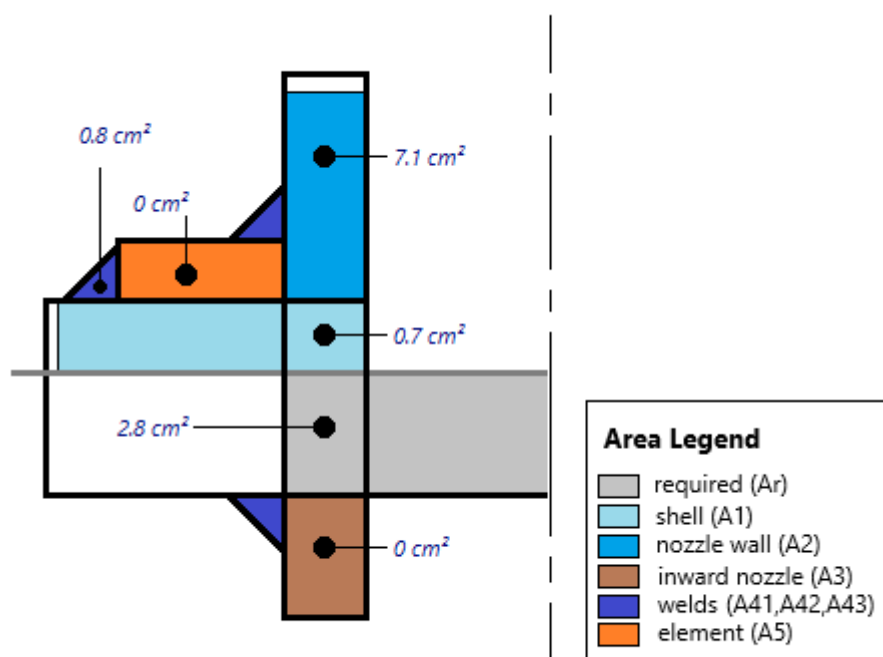


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
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Area Required	Ar	3.331	2.773	NA
Area in Shell	A1	2.541	0.699	NA
Area in Nozzle Wall	A2	7.188	7.099	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.833	0.833	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.562	8.631	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (52.8 \cdot 9.55 \cdot 1 + 2 \cdot 15.8 \cdot 9.55 \cdot 1 (1 - 0.83))$
 $= 2.773 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 53.5 (1 \cdot 11 - 1 \cdot 9.55) - 2 \cdot 15.8$
 $(1 \cdot 11 - 1 \cdot 9.55) \cdot (1 - 0.83)$
 $= 0.699 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 27.5) (15.8 - 0.28) 0.83$
 $= 7.099 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$
 $= 0.833 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.2756 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 6.7369 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 6.7369 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 1.8448 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(6.74, 2.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.28, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 15.7,	Allowable	: 115.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 271.8 N./mm ²	Passed
Occasional	: 0.2,	Allowable	: 152.9 N./mm ²	Passed
Shear	: 11.8,	Allowable	: 80.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: N4-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (2.77 - 0.7 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 11 - 9.55)) \cdot 138)$
 $= 3453.76 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (7.1 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138$
 $= 11161.83 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (7.1 + 0 + 0.83 + (2.89)) \cdot 138$
 $= 15228.60 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (7.1 + 0 + 0.83 + 0 + (2.89)) \cdot 138$
 $= 15228.60 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115$
 $= 7611. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115$
 $= 13936. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 84.3 \cdot (12 - 1) \cdot 0.74 \cdot 138$
 $= 15173. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (7611 + 13936) = 21547 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (7611 + 0 + 15173 + 0) = 22784 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (7611 + 15173 + 0) = 22784 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 21547 Kgf, must exceed W = 3453 Kgf or W1 = 11161 Kgf
 Path 2-2 = 22783 Kgf, must exceed W = 3453 Kgf or W2 = 15228 Kgf
 Path 3-3 = 22783 Kgf, must exceed W = 3453 Kgf or W3 = 15228 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 0.3945 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 172.3946 mm.

Input Echo, WRC297 Item 1, Description: N4-2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	4506.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.0000	mm.
Nozzle Diameter	50.800	mm.
Nozzle Thickness	16.7640	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-182 F316L	
Nozzle Cold S.I. Allowable	SNmc	115.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	115.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	3.51	bars
Radial Load	P	458.90	Kgf
Circumferential Shear	Vc	331.43	Kgf
Longitudinal Shear	Vl	331.43	Kgf
Circumferential Moment	Mc	50.97	Kg-m.
Longitudinal Moment	Ml	50.97	Kg-m.
Torsional Moment	Mt	58.61	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.030) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(2259.5(12.0 - 1.0))}$$

$$= 604.583 \text{ mm.}$$

WRC 297 Curve Access Parameters:



Vessel Mean Diameter (D) = 4519.000 mm.
 Nozzle Outside Diameter (d) = 84.328 mm.
 Vessel Thickness used (T) = 11.000 mm.
 Nozzle Thickness used (t) = 15.764 mm.
 T / t = 0.698
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.378$$

Nr/P = 0.169
 Mr/P = 0.351
 M0/P = 0.111
 N0/P = 0.191
 MrD/Mc = 0.306
 NrDL/Mc = 0.101
 M0d/Mc = 0.108
 N0DL/Mc = -0.070
 MrD/Ml = 0.282
 NrDL/Ml = 0.106
 M0D/Ml = 0.100
 N0DL/Ml = -0.007

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-24	24	-24	24
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	71	72	71	72
Outplane Stress Summary	10	119	68	60

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-6	-6	-6	-6
Inplane Bending (P)	-78	78	-78	78
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-82	82	82	-82
Inplane Pressure Stress	35	35	35	35
Inplane Stress Summary	-136	185	39	30

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	148	186	68	60



Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-24	24	-24	24
Outplane Membrane (Mc)	3	3	-3	-3
Outplane Bending (Mc)	-31	31	31	-31
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	35	35	35	35
Outplane Stress Summary	-24	88	32	18

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-6	-6	-6	-6
Inplane Bending (P)	-78	78	-78	78
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-90	90	90	-90
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	71	72	71	72
Inplane Stress Summary	-107	229	82	59

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	108	229	82	60

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	-6	-5	-6	-6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1



Axial Bending (P)	-28	28	-28	28
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-32	32	32	-32
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-72	50	12	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3

Shear Stress Summary	4	4	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	73	57	19	12
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	3	3	-3	-3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0

Hoop Stress Summary	-3	-2	-9	-9
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-28	28	-28	28
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-36	36	36	-36
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-76	54	16	1
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3

Shear Stress Summary	1	1	4	4
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	76	57	28	14
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	72	71	72	71	72	71	72
Circ. Pl (SUS)	-7	-7	-7	-7	-10	-10	-2	-2
Circ. Q (SUS)	-53	53	5	-5	-168	168	12	-12
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	-11	-11	-1	-1	-4	-4	-10	-10
Long. Q (SUS)	-160	160	4	-4	-55	55	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	71.0	72.0	71.0	72.0	71.0	72.0	71.0	72.0
Pm+Pl (SUS)	64.1	65.1	64.1	65.1	61.1	62.1	69.1	70.1
Pm+Pl+Q (Total)	147.5	184.5	69.1	60.1	107.0	230.0	81.7	58.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	72.00	138.00	Passed
Pm+Pl (SUS)	70.09	207.00	Passed
Pm+Pl+Q (TOTAL)	230.03	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	0	0	0	0	0	0	0
Circ. Pl (SUS)	-7	-7	-7	-7	-4	-4	-10	-10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-11	-11	9	9	-11	-11	9	9
Long. Q (SUS)	-60	60	4	-4	-64	64	8	-8
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	11.2	11.2	16.1	16.1	11.1	11.1	19.1	19.1
Pm+Pl+Q (Total)	71.2	56.6	20.4	12.6	75.1	57.1	28.2	13.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	0.00	115.00	Passed
Pm+Pl (SUS)	19.10	172.50	Passed
Pm+Pl+Q (TOTAL)	75.06	344.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K1-2"

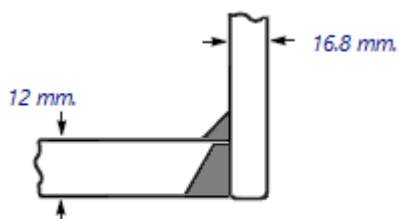
From: 50

Pressure for Reinforcement Calculations	P	3.507	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4506.00	mm.
Design Length of Section	L	3200.0000	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.51 \cdot 2254 / (138 \cdot 1 - 0.6 \cdot 3.51)$
 $= 5.7369$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.51 / (115 \cdot 1)] - 1))$
 $= 0.0806$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2756 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	106.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	53.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

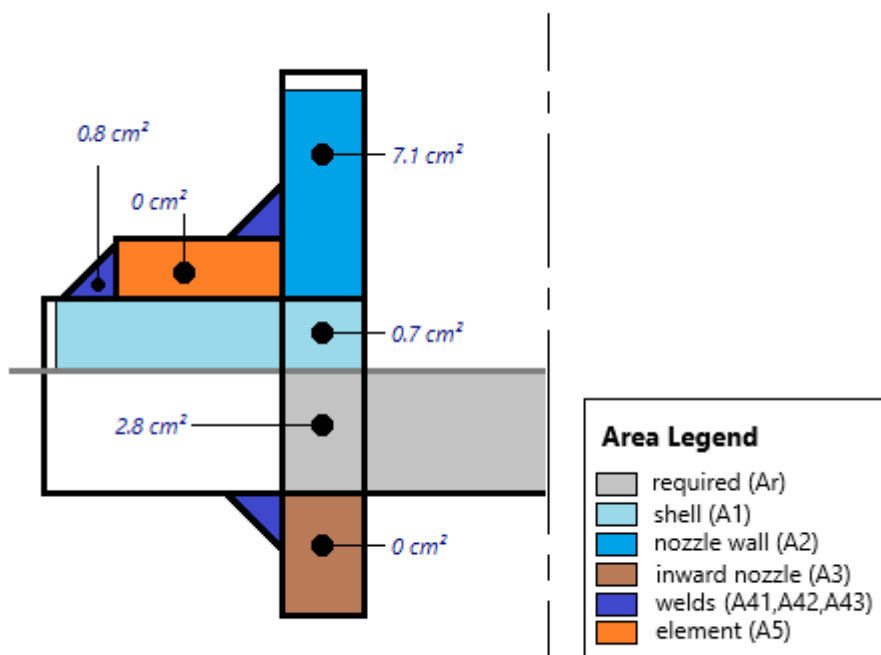


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
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Area Required	Ar	3.331	2.773	NA
Area in Shell	A1	2.541	0.699	NA
Area in Nozzle Wall	A2	7.188	7.099	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.833	0.833	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.562	8.631	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(52.8 \cdot 9.55 \cdot 1 + 2 \cdot 15.8 \cdot 9.55 \cdot 1(1-0.83))$
 $= 2.773 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 53.5 (1 \cdot 11 - 1 \cdot 9.55) - 2 \cdot 15.8$
 $(1 \cdot 11 - 1 \cdot 9.55) \cdot (1 - 0.83)$
 $= 0.699 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 27.5) (15.8 - 0.28) 0.83$
 $= 7.099 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$
 $= 0.833 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.2756 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 6.7369 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 6.7369 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 1.8448 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(6.74, 2.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.28, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (2.77 - 0.7 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 11 - 9.55)) \cdot 138)$
 $= 3453.76 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (7.1 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138$
 $= 11161.83 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (7.1 + 0 + 0.83 + (2.89)) \cdot 138$
 $= 15228.60 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (7.1 + 0 + 0.83 + 0 + (2.89)) \cdot 138$
 $= 15228.60 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115$
 $= 7611. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115$
 $= 13936. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 84.3 \cdot (12 - 1) \cdot 0.74 \cdot 138$
 $= 15173. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (7611 + 13936) = 21547 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (7611 + 0 + 15173 + 0) = 22784 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (7611 + 15173 + 0) = 22784 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 21547	Kgf, must exceed W = 3453 Kg	or W1 = 11161 Kg
Path 2-2 = 22783	Kgf, must exceed W = 3453 Kg	or W2 = 15228 Kg
Path 3-3 = 22783	Kgf, must exceed W = 3453 Kg	or W3 = 15228 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 0.3945 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 172.3946 mm.

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Input, Nozzle Desc: K2-2"

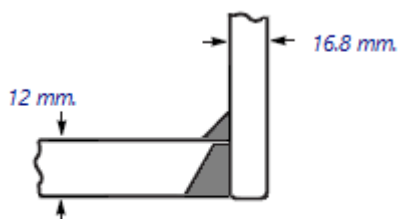
From: 50

Pressure for Reinforcement Calculations	P	3.541	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4506.00	mm.
Design Length of Section	L	925.5002	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.54 \cdot 2254 / (138 \cdot 1 - 0.6 \cdot 3.54)$
 $= 5.7923$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.54 / (115 \cdot 1)] - 1))$
 $= 0.0814$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2756 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	106.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	53.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

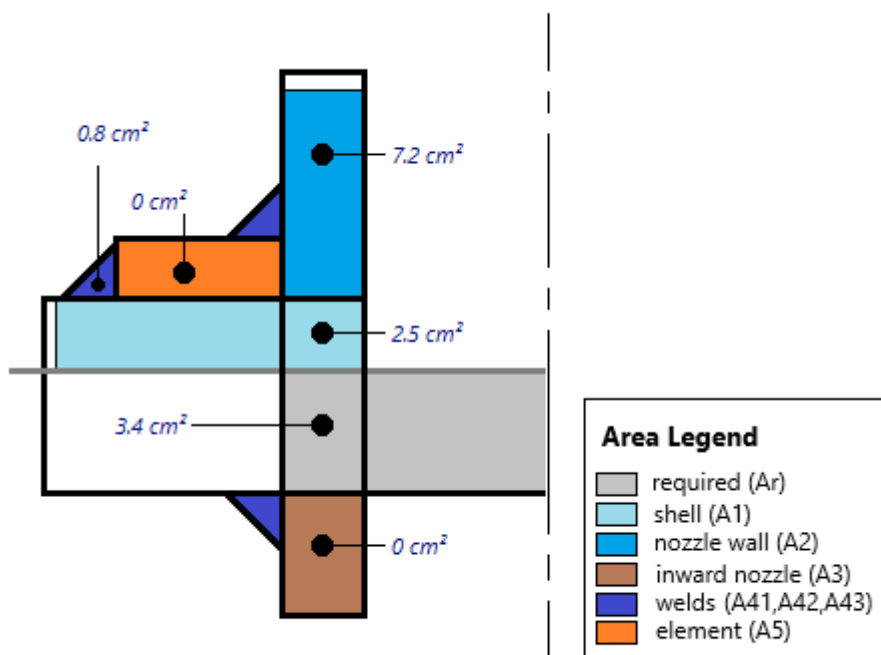


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
--------------------------	------	----------	-------

Area Required	Ar	3.363	1.654	NA
Area in Shell	A1	2.514	2.560	NA
Area in Nozzle Wall	A2	7.188	7.099	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.833	0.833	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.535	10.492	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (52.8 \cdot 5.79 \cdot 1 + 2 \cdot 15.8 \cdot 5.79 \cdot 1 \cdot (1 - 0.83))$
 $= 3.363 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 53.5 (1 \cdot 11 - 1 \cdot 5.79) - 2 \cdot 15.8$
 $(1 \cdot 11 - 1 \cdot 5.79) \cdot (1 - 0.83)$
 $= 2.514 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 27.5) (15.8 - 0.081) 0.83$
 $= 7.188 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$
 $= 0.833 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.2756 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 6.7923 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 6.7923 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 1.8448 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(6.79, 2.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.28, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3.36 - 2.51 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 11 - 5.79)) \cdot 138)$
 $= 3119.81 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (7.19 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138$
 $= 11287.07 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (7.19 + 0 + 0.83 + (2.89)) \cdot 138$
 $= 15353.84 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (7.19 + 0 + 0.83 + 0 + (2.89)) \cdot 138$
 $= 15353.84 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115$
 $= 7611. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115$
 $= 13936. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 84.3 \cdot (12 - 1) \cdot 0.74 \cdot 138$
 $= 15173. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (7611 + 13936) = 21547 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (7611 + 0 + 15173 + 0) = 22784 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (7611 + 15173 + 0) = 22784 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 21547	Kgf, must exceed W = 3119 Kg	or W1 = 11287 Kg
Path 2-2 = 22783	Kgf, must exceed W = 3119 Kg	or W2 = 15353 Kg
Path 3-3 = 22783	Kgf, must exceed W = 3119 Kg	or W3 = 15353 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 0.3945 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 172.3946 mm.

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Input, Nozzle Desc: K4-2"

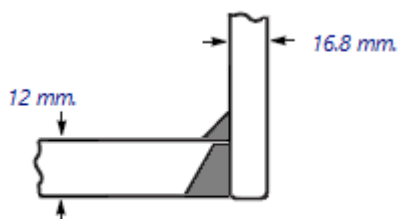
From: 50

Pressure for Reinforcement Calculations	P	3.541	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4506.00	mm.
Design Length of Section	L	925.5002	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.54 \cdot 2254 / (138 \cdot 1 - 0.6 \cdot 3.54)$
 $= 5.7923$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.54 / (115 \cdot 1)] - 1))$
 $= 0.0814$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2756 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	106.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	53.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

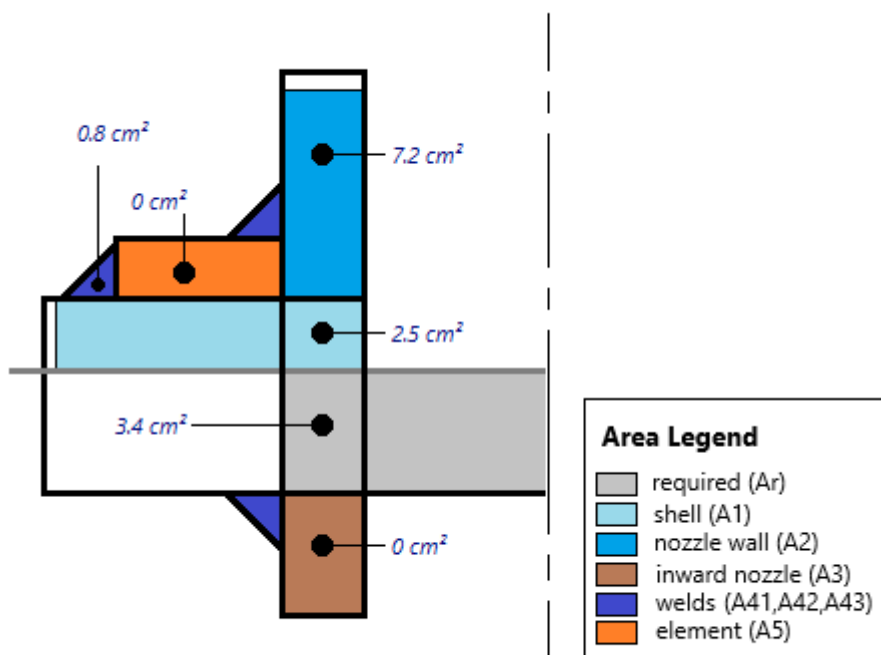


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
--------------------------	------	----------	-------



Area Required	Ar	3.363	1.654	NA
Area in Shell	A1	2.514	2.560	NA
Area in Nozzle Wall	A2	7.188	7.099	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.833	0.833	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.535	10.492	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (52.8 \cdot 5.79 \cdot 1 + 2 \cdot 15.8 \cdot 5.79 \cdot 1 \cdot (1 - 0.83))$
 $= 3.363 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 53.5(1 \cdot 11 - 1 \cdot 5.79) - 2 \cdot 15.8$
 $(1 \cdot 11 - 1 \cdot 5.79) \cdot (1 - 0.83)$
 $= 2.514 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 27.5)(15.8 - 0.081)0.83$
 $= 7.188 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$
 $= 0.833 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.2756 \text{ mm}$.
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$.
 Wall Thickness, shell/head, internal pressure $trb1 = 6.7923 \text{ mm}$.
 Wall Thickness $tb1 = \max(trb1, tr16b) = 6.7923 \text{ mm}$.
 Wall Thickness, shell/head, external pressure $trb2 = 1.8448 \text{ mm}$.
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm}$.
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm}$.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(6.79, 2.5)]$
 $= 5.8000 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.28, 5.8)$
 $= 5.8000 \text{ mm}$.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (3.36 - 2.51 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 11 - 5.79)) \cdot 138)$
 $= 3119.81 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (7.19 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138$
 $= 11287.07 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (7.19 + 0 + 0.83 + (2.89)) \cdot 138$
 $= 15353.84 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (7.19 + 0 + 0.83 + 0 + (2.89)) \cdot 138$
 $= 15353.84 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115$
 $= 7611. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115$
 $= 13936. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 84.3 \cdot (12 - 1) \cdot 0.74 \cdot 138$
 $= 15173. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (7611 + 13936) = 21547 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (7611 + 0 + 15173 + 0) = 22784 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (7611 + 15173 + 0) = 22784 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 21547	Kgf, must exceed W = 3119 Kg	or W1 = 11287 Kg
Path 2-2 = 22783	Kgf, must exceed W = 3119 Kg	or W2 = 15353 Kg
Path 3-3 = 22783	Kgf, must exceed W = 3119 Kg	or W3 = 15353 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 0.3945 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 172.3946 mm.

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Input, Nozzle Desc: K5-2"

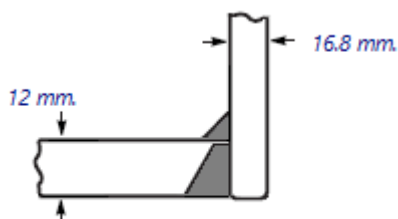
From: 50

Pressure for Reinforcement Calculations	P	3.507	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4506.00	mm.
Design Length of Section	L	3200.0000	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		900.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		135.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.51 \cdot 2254 / (138 \cdot 1 - 0.6 \cdot 3.51)$
 $= 5.7369$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.51 / (115 \cdot 1)] - 1))$
 $= 0.0806$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2756 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	106.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	53.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

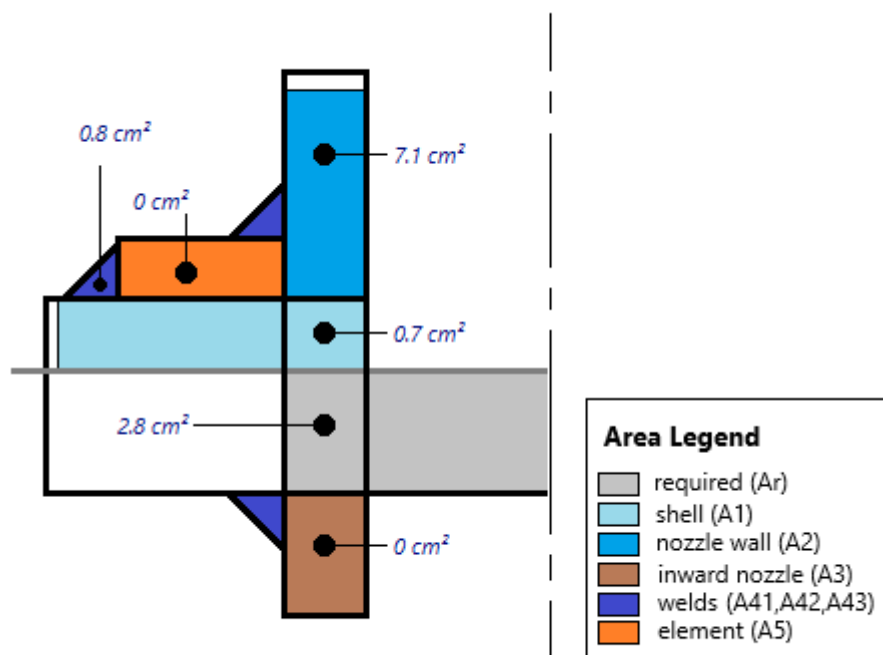


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
--------------------------	------	----------	-------

Area Required	Ar	3.331	2.773	NA
Area in Shell	A1	2.541	0.699	NA
Area in Nozzle Wall	A2	7.188	7.099	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.833	0.833	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.562	8.631	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(52.8 \cdot 9.55 \cdot 1 + 2 \cdot 15.8 \cdot 9.55 \cdot 1(1-0.83))$
 $= 2.773 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 53.5 (1 \cdot 11 - 1 \cdot 9.55) - 2 \cdot 15.8$
 $(1 \cdot 11 - 1 \cdot 9.55) \cdot (1 - 0.83)$
 $= 0.699 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 27.5) (15.8 - 0.28) 0.83$
 $= 7.099 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$
 $= 0.833 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.2756 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 6.7369 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 6.7369 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 1.8448 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(6.74, 2.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.28, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K5-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (2.77 - 0.7 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 11 - 9.55)) \cdot 138)$
 $= 3453.76 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (7.1 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138$
 $= 11161.83 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (7.1 + 0 + 0.83 + (2.89)) \cdot 138$
 $= 15228.60 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (7.1 + 0 + 0.83 + 0 + (2.89)) \cdot 138$
 $= 15228.60 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115$
 $= 7611. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115$
 $= 13936. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 84.3 \cdot (12 - 1) \cdot 0.74 \cdot 138$
 $= 15173. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (7611 + 13936) = 21547 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (7611 + 0 + 15173 + 0) = 22784 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (7611 + 15173 + 0) = 22784 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 21547	Kgf, must exceed W = 3453 Kg	or W1 = 11161 Kg
Path 2-2 = 22783	Kgf, must exceed W = 3453 Kg	or W2 = 15228 Kg
Path 3-3 = 22783	Kgf, must exceed W = 3453 Kg	or W3 = 15228 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 0.3945 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 172.3946 mm.

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Input, Nozzle Desc: K3-2:

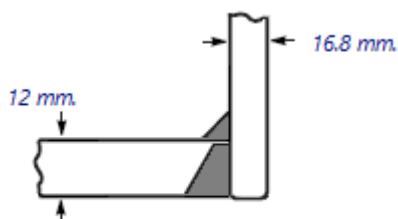
From: 50

Pressure for Reinforcement Calculations	P	3.507	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4506.00	mm.
Design Length of Section	L	3200.0000	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		900.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3-2:

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.51 \cdot 2254 / (138 \cdot 1 - 0.6 \cdot 3.51)$
 $= 5.7369$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.51 / (115 \cdot 1)] - 1))$
 $= 0.0806$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2756 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	106.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	53.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

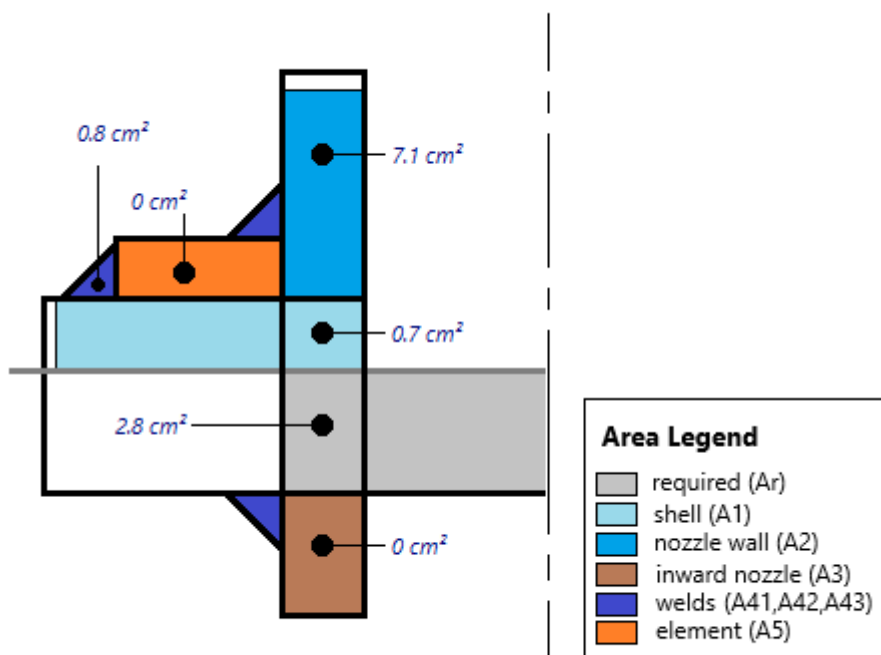


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
--------------------------	------	----------	-------

Area Required	Ar	3.331	2.773	NA
Area in Shell	A1	2.541	0.699	NA
Area in Nozzle Wall	A2	7.188	7.099	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.833	0.833	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.562	8.631	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr_1))$ per UG-37(d)
 $= 0.5(52.8 \cdot 9.55 \cdot 1 + 2 \cdot 15.8 \cdot 9.55 \cdot 1(1-0.83))$
 $= 2.773 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr_1)$
 $= 53.5 (1 \cdot 11 - 1 \cdot 9.55) - 2 \cdot 15.8$
 $(1 \cdot 11 - 1 \cdot 9.55) \cdot (1 - 0.83)$
 $= 0.699 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr_2$
 $= (2 \cdot 27.5) (15.8 - 0.28) 0.83$
 $= 7.099 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr_2 + (Wi-can/0.707)^2 \cdot fr_2$
 $= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$
 $= 0.833 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.2756 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 6.7369 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 6.7369 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 1.8448 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(6.74, 2.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.28, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K3-2:

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (2.77 - 0.7 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 11 - 9.55)) \cdot 138)$
 $= 3453.76 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (7.1 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138$
 $= 11161.83 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (7.1 + 0 + 0.83 + (2.89)) \cdot 138$
 $= 15228.60 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (7.1 + 0 + 0.83 + 0 + (2.89)) \cdot 138$
 $= 15228.60 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115$
 $= 7611. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115$
 $= 13936. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 84.3 \cdot (12 - 1) \cdot 0.74 \cdot 138$
 $= 15173. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (7611 + 13936) = 21547 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (7611 + 0 + 15173 + 0) = 22784 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (7611 + 15173 + 0) = 22784 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 21547	Kgf, must exceed W = 3453 Kg	or W1 = 11161 Kg
Path 2-2 = 22783	Kgf, must exceed W = 3453 Kg	or W2 = 15228 Kg
Path 3-3 = 22783	Kgf, must exceed W = 3453 Kg	or W3 = 15228 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 0.3945 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 172.3946 mm.

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Input, Nozzle Desc: K6-2"

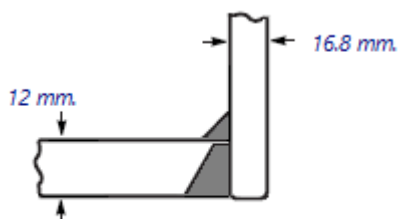
From: 50

Pressure for Reinforcement Calculations	P	3.541	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4506.00	mm.
Design Length of Section	L	925.5002	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.54 \cdot 2254 / (138 \cdot 1 - 0.6 \cdot 3.54)$
 $= 5.7923$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.54 / (115 \cdot 1)] - 1))$
 $= 0.0814$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2756 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	106.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	53.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

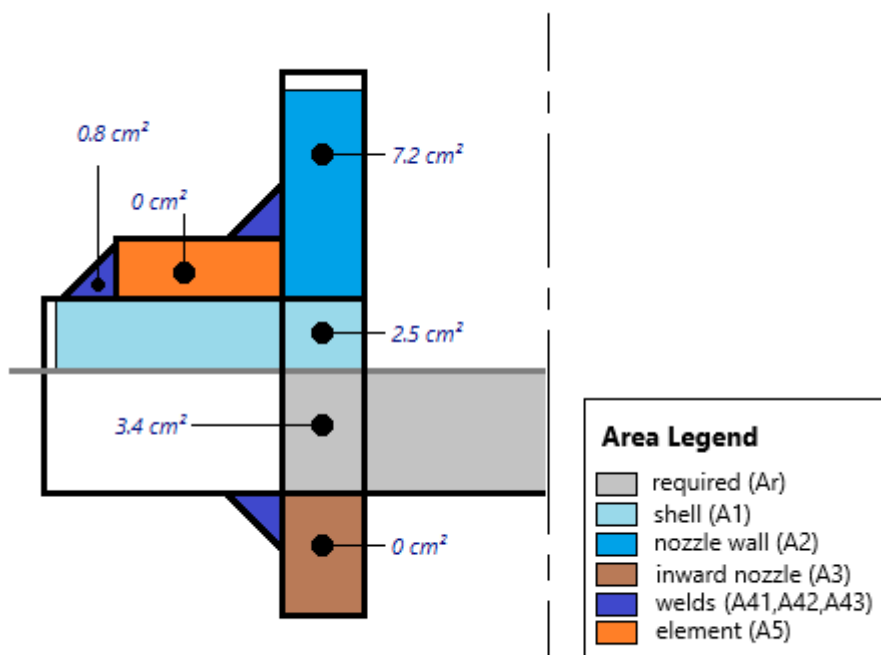


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
--------------------------	------	----------	-------

Area Required	Ar	3.363	1.654	NA
Area in Shell	A1	2.514	2.560	NA
Area in Nozzle Wall	A2	7.188	7.099	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.833	0.833	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.535	10.492	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (52.8 \cdot 5.79 \cdot 1 + 2 \cdot 15.8 \cdot 5.79 \cdot 1 \cdot (1 - 0.83))$
 $= 3.363 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 53.5(1 \cdot 11 - 1 \cdot 5.79) - 2 \cdot 15.8$
 $(1 \cdot 11 - 1 \cdot 5.79) \cdot (1 - 0.83)$
 $= 2.514 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 27.5)(15.8 - 0.081)0.83$
 $= 7.188 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$
 $= 0.833 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.2756 \text{ mm}$.
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$.
 Wall Thickness, shell/head, internal pressure $trb1 = 6.7923 \text{ mm}$.
 Wall Thickness $tb1 = \max(trb1, tr16b) = 6.7923 \text{ mm}$.
 Wall Thickness, shell/head, external pressure $trb2 = 1.8448 \text{ mm}$.
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm}$.
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm}$.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(6.79, 2.5)]$
 $= 5.8000 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.28, 5.8)$
 $= 5.8000 \text{ mm}$.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (3.36 - 2.51 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 11 - 5.79)) \cdot 138)$
 $= 3119.81 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (7.19 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138$
 $= 11287.07 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (7.19 + 0 + 0.83 + (2.89)) \cdot 138$
 $= 15353.84 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (7.19 + 0 + 0.83 + 0 + (2.89)) \cdot 138$
 $= 15353.84 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115$
 $= 7611. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115$
 $= 13936. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 84.3 \cdot (12 - 1) \cdot 0.74 \cdot 138$
 $= 15173. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (7611 + 13936) = 21547 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (7611 + 0 + 15173 + 0) = 22784 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (7611 + 15173 + 0) = 22784 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 21547 Kgf, must exceed W = 3119 Kgf	or W1 = 11287 Kgf
Path 2-2 = 22783 Kgf, must exceed W = 3119 Kgf	or W2 = 15353 Kgf
Path 3-3 = 22783 Kgf, must exceed W = 3119 Kgf	or W3 = 15353 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 0.3945 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 172.3946 mm.

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Input, Nozzle Desc: M1-24"

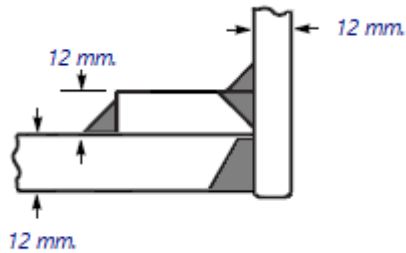
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Pressure for Reinforcement Calculations	P	3.507	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4506.00	mm.
Design Length of Section	L	3200.0000	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2050.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	930.0000	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		148.2000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.51 \cdot 2254 / (138 \cdot 1 - 0.6 \cdot 3.51)$
 $= 5.7369 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.51 \cdot 306 / (138 \cdot 1 - 0.6 \cdot 3.51)$
 $= 0.7783 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.2090 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1223.2001 mm.
Parallel to Vessel Wall, opening length	d	611.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

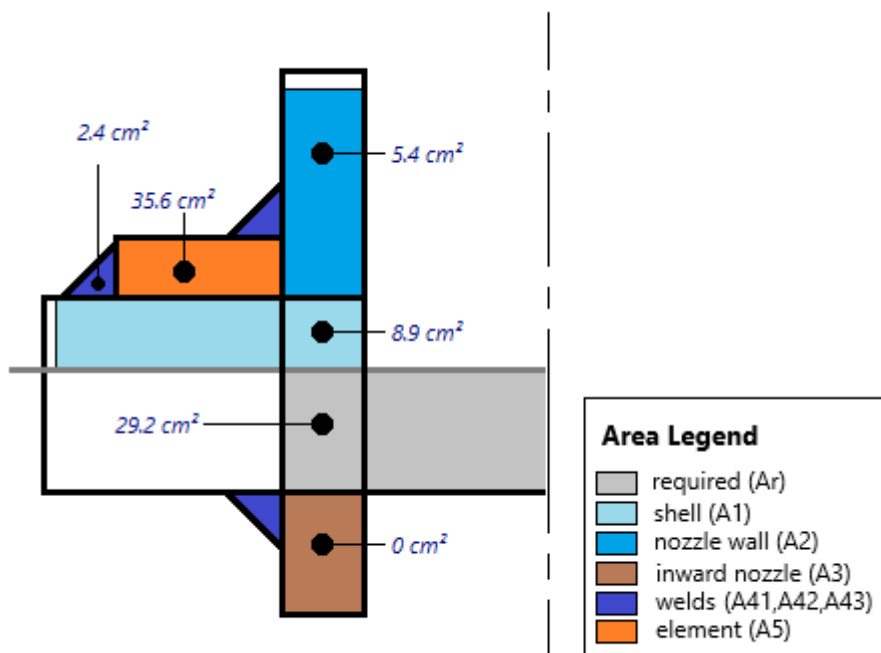


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	35.087	29.211	NA
Area in Shell	A1	32.189	8.853	NA
Area in Nozzle Wall	A2	5.622	5.385	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.440	2.440	NA
Area in Element	A5	35.568	35.568	NA
TOTAL AREA AVAILABLE	Atot	75.819	52.246	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 738.0418 12.0000 mm.
 Based on given Pad Diameter: 930.0000 4.2284 mm.
 Based on Shell or Nozzle Thickness: 738.0418 12.0000 mm.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(612 \cdot 9.55 \cdot 1 + 2 \cdot 11 \cdot 9.55 \cdot 1(1 - 1))$
 $= 29.211 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 612(1 \cdot 11 - 1 \cdot 9.55) - 2 \cdot 11$
 $(1 \cdot 11 - 1 \cdot 9.55) \cdot (1 - 1)$
 $= 8.853 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 27.5) \cdot (11 - 1.21) \cdot 1$

= 5.385 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 = $10^2 \cdot 1 + (0)^2 \cdot 1 + 12^2 \cdot 1$
 = 2.440 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 = $(930 - 634) \cdot 12 \cdot 1$
 = 35.568 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 2.2090$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 2.2090 mm.

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 12$, $t_r = 0.78$, $c = 1$ mm., $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 12$, $t_r = 0.78$, $c = 1$ mm., $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 12$, $t_r = 0.78$, $c = 1$ mm., $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 12$, $t_r = 5.74$, $c = 1$ mm., $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $t_g = 12$, $t_r = 5.74$, $c = 1$ mm., $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.51/19.60 = 0.179$

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 11.0000$ mm.
 Intermediate Calc. for pad/shell Welds $T_{minPad} = 11.0000$ mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$5.5000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_r \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (29.2 - 8.85 + 2 \cdot 11 \cdot 1 \cdot (1 \cdot 11 - 9.55)) \cdot 138)$
 $= 29095.68 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - \text{Can} / .707)^2 \cdot f_r) \cdot S_v$
 $= (5.39 + 35.6 + 2.44 - 0 \cdot 1) \cdot 138$
 $= 61061.96 \text{ Kgf}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_r)) \cdot S_v$
 $= (5.39 + 0 + 1 + (2.42)) \cdot 138$
 $= 12390.34 \text{ Kgf}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_r)) \cdot S$
 $= (5.39 + 0 + 2.44 + 35.6 + (2.42)) \cdot 138$
 $= 64467.34 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 634 \cdot 10 \cdot 0.49 \cdot 138$
 $= 68625. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 930 \cdot 12 \cdot 0.49 \cdot 138$
 $= 120873. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 311) \cdot (12 - 1) \cdot 0.7 \cdot 138$
 $= 105967. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 634 \cdot 12 \cdot 0.74 \cdot 138$
 $= 124365. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 634 \cdot (12 - 1) \cdot 0.74 \cdot 138$
 $= 114001. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:



$$= (\text{SPEW} + \text{SNW}) = (120873 + 105967) = 226840 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (68625 + 124365 + 114001 + 0) = 306991 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (120873 + 114001 + 0) = 234875 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 226840 Kgf, must exceed W = 29095 Kgf or W1 = 61061 Kgf
Path 2-2 = 306991 Kgf, must exceed W = 29095 Kgf or W2 = 12390 Kgf
Path 3-3 = 234874 Kgf, must exceed W = 29095 Kgf or W3 = 64467 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 22.3843 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 384.3843 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 * t_{nom} / R_f * (1 - R_f / R_o))$ 1.931 %

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 611.60 mm.

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Input, Nozzle Desc: N2-42"

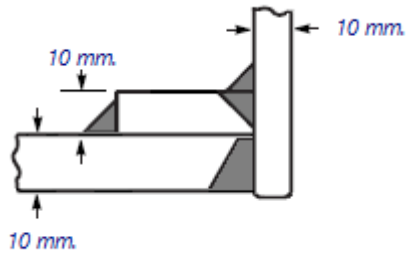
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Pressure for Reinforcement Calculations	P	3.507	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	4506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		42.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	375.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1949.9999	mm.
Thickness of Pad	te	10.0000	mm.
Weld leg size between Pad and Shell	Wp	10.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm.
Reinforcing Pad Width		431.5999	mm.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N2-42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	42.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.51 \cdot 4508 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.51)$
 $= 5.1544$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.51 \cdot 534 / (138 \cdot 1 - 0.6 \cdot 3.51)$
 $= 1.3601$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.6979 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	2137.5999	mm.
Parallel to Vessel Wall, opening length	d	1068.7999	mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

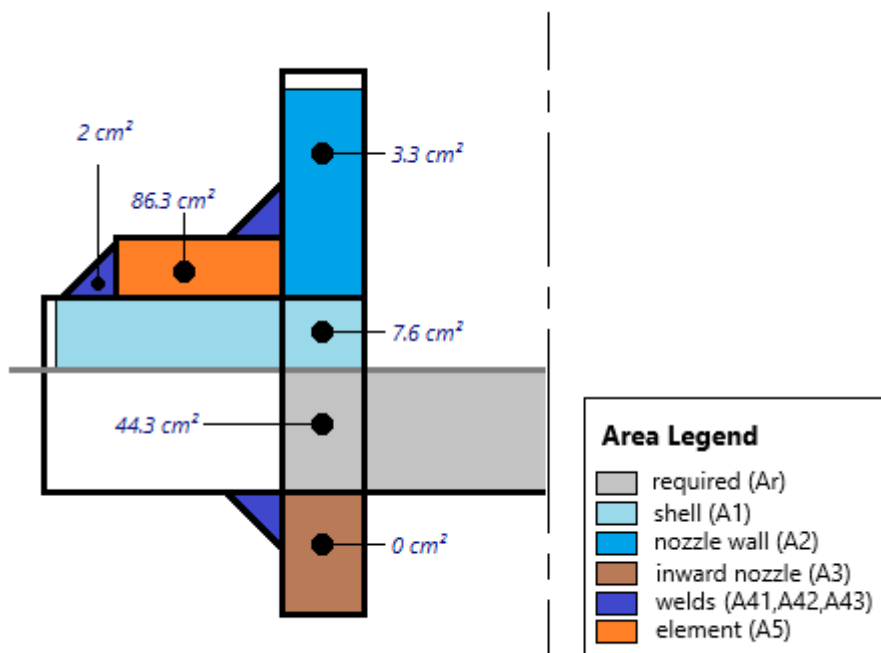


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	55.090	44.282	NA
Area in Shell	A1	41.102	7.629	NA
Area in Nozzle Wall	A2	3.438	3.286	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.000	2.000	NA
Area in Element	A5	86.320	86.320	NA
TOTAL AREA AVAILABLE	Atot	132.859	99.234	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1400.4720 10.0000 mm.
 Based on given Pad Diameter: 1949.9999 3.6338 mm.
 Based on Shell or Nozzle Thickness: 1400.4720 10.0000 mm.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (1069 \cdot 8.29 \cdot 1 + 2 \cdot 9 \cdot 8.29 \cdot 1 (1 - 1))$
 $= 44.282 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 1069 (1 \cdot 9 - 1 \cdot 8.29) - 2 \cdot 9$
 $(1 \cdot 9 - 1 \cdot 8.29) \cdot (1 - 1)$
 $= 7.629 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 22.5) \cdot (9 - 1.7) \cdot 1$

= 3.286 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr3 + (W_i - c_{an}/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 = $10^2 \cdot 1 + (0)^2 \cdot 1 + 10^2 \cdot 1$
 = 2.000 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 = $(1950 - 1087) \cdot 10 \cdot 1$
 = 86.320 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.6979 mm.
Wall Thickness per UG16(b),	tr16b = 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 6.7262 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 6.7262 mm.
Wall Thickness, shell/head, external pressure	trb2 = 1.8441 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm.
Wall Thickness per table UG-45	tb3 = 9.3312 mm.

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[9.33, max(6.73, 2.5)]
 = 6.7262 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(2.7, 6.73)
 = 6.7262 mm.

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	39.7,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	305.3 N./mm ²	Passed
Occasional	:	10.3,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	18.6,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, tg = 10, tr = 1.36, c = 1 mm., E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, tg = 10, tr = 1.36, c = 1 mm., E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 10, tr = 1.36, c = 1 mm., E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 10, tr = 5.15, c = 1 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.57$, Temp. Reduction = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: D

Govrn. thk, tg = 10, tr = 5.15, c = 1 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.57$, Temp. Reduction = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.51/19.60 = 0.179$

Weld Size Calculations, Description: N2-42"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 10.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$5.0000 = 0.5 \cdot T_{\text{minPad}}$	$7.0700 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (44.3 - 7.63 + 2 \cdot 9 \cdot 1 \cdot (1 \cdot 9 - 8.29)) \cdot 138)$
 $= 51758.46 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - C_{an} / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (3.29 + 86.3 + 2 - 0 \cdot 1) \cdot 138$
 $= 128906.26 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (3.29 + 0 + 1 + (1.62)) \cdot 138$
 $= 8310.75 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (3.29 + 0 + 2 + 86.3 + (1.62)) \cdot 138$
 $= 131185.89 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 1087 \cdot 10 \cdot 0.49 \cdot 138$
 $= 117711. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$



$$= (3.14/2.0) \cdot 1950 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 211203. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 539) \cdot (10 - 1) \cdot 0.7 \cdot 138$$

$$= 150089. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 1087 \cdot 10 \cdot 0.74 \cdot 138$$

$$= 177767. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1087 \cdot (10 - 1) \cdot 0.74 \cdot 138$$

$$= 159990. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (211203 + 150089) = 361292 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (117711 + 177767 + 159990 + 0) = 455468 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (211203 + 159990 + 0) = 371194 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 361292 Kgf, must exceed W = 51758 Kgf	or W1 = 128906 Kgf
Path 2-2 = 455468 Kgf, must exceed W = 51758 Kgf	or W2 = 8310 Kgf
Path 3-3 = 371193 Kgf, must exceed W = 51758 Kgf	or W3 = 131185 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 36.4985 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 421.4987 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.929 %

Input Echo, WRC107/537 Item 1, Description: N2-42" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	1.0000 mm.
Vessel Diameter	Dv	8110.800 mm.
Vessel Thickness	Tv	10.000 mm.
Design Temperature	T1	110.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow



Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.0000	mm.
Nozzle Diameter	Dn	1066.800	mm.
Nozzle Thickness	Tn	10.000	mm.
Nozzle Material		SA-516 70	
Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

Thickness of Reinforcing Pad	Tpad	10.000	mm.
Diameter of Reinforcing Pad	Dpad	1950.000	mm.

Design Internal Pressure	Dp	3.507	bars
Include Pressure Thrust	No		

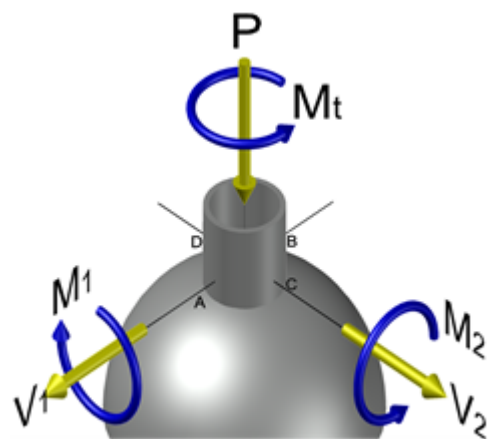
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	8030.8	Kgf
Longitudinal Shear (SUS)	(V1) V1	5353.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	5353.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	15734.3	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	15734.3	Kg-m.
Torsional Moment (SUS)	Mt	22477.6	Kg-m.

Use Interactive Control	No		
WRC107 Version	Version	March	1979

Include Pressure Stress Indices per Div. 2	No		
Compute Pressure Stress per WRC-368	No		
Local Loads applied at end of Nozzle/Attachment	No		

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 1086.8 + 2 \cdot 1.65 \cdot \sqrt{(4060.9 (10.0 - 1.0))} \\
 &= 1717.679 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	8030.8	Kgf
Circumferential Shear	(VC) V2	5353.9	Kgf
Longitudinal Shear	(VL) V1	5353.9	Kgf
Circumferential Moment	(MC) M1	15734.3	Kg-m.
Longitudinal Moment	(ML) M2	15734.3	Kg-m.
Torsional Moment	MT	22477.6	Kg-m.



Unitless Prm: U = 1.96 TAU = 50.00 (59.88) RHO = 4.00 (2.11)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 9	0.00788	(A,B,C,D)
M(x) / P	SP 9	0.00480	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 9	0.00832	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 9	0.00504	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 9	0.00832	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 9	0.00504	(A,B,C,D)
N(y) * T / P	SP 9	0.11901	(A,B,C,D)
M(y) / P	SP 9	0.00772	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 9	0.10611	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 9	0.00984	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 9	0.10611	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 9	0.00984	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7
Rad.	Bend. P	-6.3	6.3	-6.3	6.3	-6.3	6.3	-6.3	6.3
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-12.8	-12.8	12.8	12.8
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-46.5	46.5	46.5	-46.5
Rad.	Memb. ML	-12.8	-12.8	12.8	12.8	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-46.5	46.5	46.5	-46.5	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-67.3	38.3	51.3	-29.1	-67.3	38.3	51.3	-29.1
Tang.	Memb. P	-26.0	-26.0	-26.0	-26.0	-26.0	-26.0	-26.0	-26.0
Tang.	Bend. P	-10.1	10.1	-10.1	10.1	-10.1	10.1	-10.1	10.1
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-163.2	-163.2	163.2	163.2
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-90.8	90.8	90.8	-90.8
Tang.	Memb. ML	-163.2	-163.2	163.2	163.2	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-90.8	90.8	90.8	-90.8	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-290.1	-88.2	217.9	56.5	-290.1	-88.2	217.9	56.5
	Shear VC	1.6	1.6	-1.6	-1.6	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
	Shear MT	6.3	6.3	6.3	6.3	6.3	6.3	6.3	6.3
	Tot. Shear	7.9	7.9	4.6	4.6	4.6	4.6	7.9	7.9
	Str. Int.	290.3	127.5	218.1	86.1	290.2	126.8	218.3	87.1

Unitless Prm: U = 5.10 TAU = 0.00 (107.83) RHO = 0.00 (1.00)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.02175	(A,B,C,D)
M(x) / P	SR 2	0.00990	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.01053	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.01053	(A,B,C,D)
N(y) * T / P	SR 2	0.00650	(A,B,C,D)
M(y) / P	SR 2	0.00300	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.00507	(A,B,C,D)



M(y) * SQRT(Rm * T) / MC SR 3 0.00317 (A,B,C,D)
 N(y) * T * SQRT(Rm * T) / ML SR 3 0.00507 (A,B,C,D)
 M(y) * SQRT(Rm * T) / ML SR 3 0.00317 (A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-21.1	-21.1	-21.1	-21.1	-21.1	-21.1	-21.1	-21.1
Rad. Bend. P		-57.8	57.8	-57.8	57.8	-57.8	57.8	-57.8	57.8
Rad. Memb. MC		0.0	0.0	0.0	0.0	-168.5	-168.5	168.5	168.5
Rad. Bend. MC		0.0	0.0	0.0	0.0	-629.4	629.4	629.4	-629.4
Rad. Memb. ML		-168.5	-168.5	168.5	168.5	0.0	0.0	0.0	0.0
Rad. Bend. ML		-629.4	629.4	629.4	-629.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-876.8	497.4	719.0	-424.2	-876.8	497.4	719.0	-424.2
Tang. Memb. P		-6.3	-6.3	-6.3	-6.3	-6.3	-6.3	-6.3	-6.3
Tang. Bend. P		-17.5	17.5	-17.5	17.5	-17.5	17.5	-17.5	17.5
Tang. Memb. MC		0.0	0.0	0.0	0.0	-50.5	-50.5	50.5	50.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-189.8	189.8	189.8	-189.8
Tang. Memb. ML		-50.5	-50.5	50.5	50.5	0.0	0.0	0.0	0.0
Tang. Bend. ML		-189.8	189.8	189.8	-189.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-264.1	150.5	216.5	-128.1	-264.1	150.5	216.5	-128.1
Shear VC		1.9	1.9	-1.9	-1.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.9	-1.9	1.9	1.9
Shear MT		4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1
Tot. Shear		6.0	6.0	2.2	2.2	2.2	2.2	6.0	6.0
Str. Int.		876.9	497.6	719.0	424.2	876.8	497.5	719.1	424.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		37.4	37.4	37.4	37.4	37.4	37.4	37.4	37.4
Rad. Pl (SUS)		-14.5	-14.5	11.1	11.1	-14.5	-14.5	11.1	11.1
Rad. Q (SUS)		-52.8	52.8	40.2	-40.2	-52.8	52.8	40.2	-40.2
Long. Pm (SUS)		37.4	37.4	37.4	37.4	37.4	37.4	37.4	37.4
Long. Pl (SUS)		-189.1	-189.1	137.2	137.2	-189.1	-189.1	137.2	137.2
Long. Q (SUS)		-100.9	100.9	80.7	-80.7	-100.9	100.9	80.7	-80.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.6	1.6	-1.6	-1.6	-1.6	-1.6	1.6	1.6
Shear Q (SUS)		6.3	6.3	6.3	6.3	6.3	6.3	6.3	6.3
Pm (SUS)		37.4	37.4	37.4	37.4	37.4	37.4	37.4	37.4
Pm+Pl (SUS)		174.7	174.7	174.7	174.7	174.7	174.7	174.7	174.7
Pm+Pl+Q (Total)		252.9	127.5	255.5	94.2	252.7	126.8	255.7	94.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	37.44	138.00	Passed
Pm+Pl (SUS)	174.67	207.00	Passed
Pm+Pl+Q (TOTAL)	255.74	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	79.0	79.0	79.0	79.0	79.0	79.0	79.0	79.0
Rad.	Pl (SUS)	-189.7	-189.7	147.4	147.4	-189.7	-189.7	147.4	147.4
Rad.	Q (SUS)	-687.1	687.1	571.6	-571.6	-687.1	687.1	571.6	-571.6
Long.	Pm (SUS)	79.0	79.0	79.0	79.0	79.0	79.0	79.0	79.0
Long.	Pl (SUS)	-56.8	-56.8	44.2	44.2	-56.8	-56.8	44.2	44.2
Long.	Q (SUS)	-207.3	207.3	172.3	-172.3	-207.3	207.3	172.3	-172.3
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.9	1.9	-1.9	-1.9	-1.9	-1.9	1.9	1.9
Shear	Q (SUS)	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1
	Pm (SUS)	79.0	79.0	79.0	79.0	79.0	79.0	79.0	79.0
	Pm+Pl (SUS)	132.9	132.9	226.5	226.5	132.9	132.9	226.5	226.5
	Pm+Pl+Q (Total)	797.9	576.6	798.1	345.2	797.8	576.5	798.1	345.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	79.03	138.00	Passed
Pm+Pl (SUS)	226.46	207.00	Failed
Pm+Pl+Q (TOTAL)	798.12	414.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K7-2"

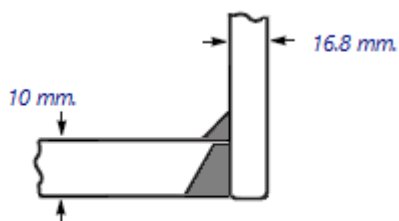
From: 70

Pressure for Reinforcement Calculations	P	3.507	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	4506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	750.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K7-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.51 \cdot 4508 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.51)$
 $= 5.1544 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.51 / (115 \cdot 1)] - 1))$
 $= 0.0806 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5345 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	115.5970 mm.
Parallel to Vessel Wall, opening length	d	57.7985 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 115/138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 115/138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

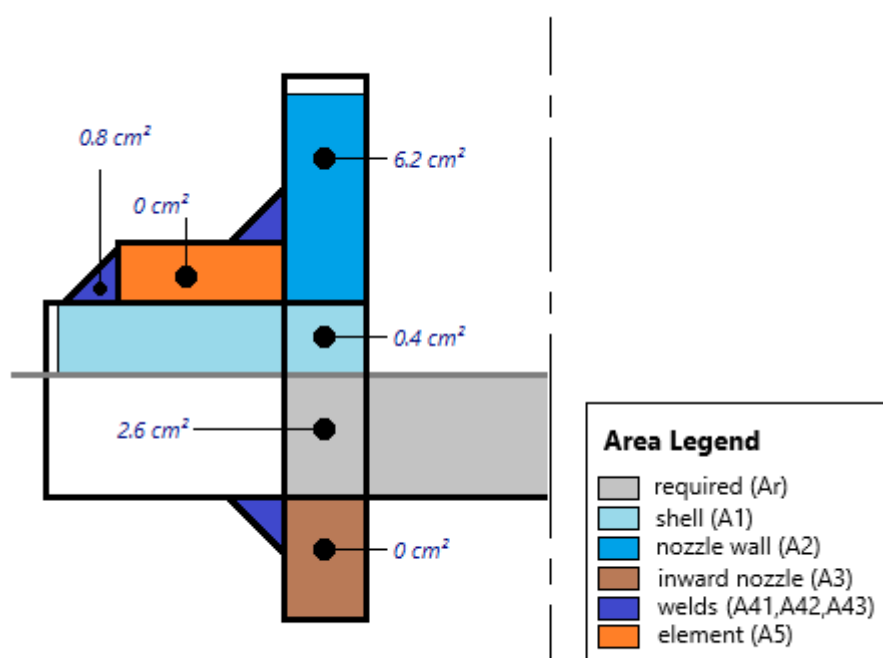


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3.250	2.612	NA
Area in Shell	A1	2.021	0.375	NA
Area in Nozzle Wall	A2	6.400	6.214	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.833	0.833	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	9.253	7.423	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 66.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(57.8 \cdot 8.29 \cdot 1 + 2 \cdot 15.8 \cdot 8.29 \cdot 1(1 - 0.83))$
 $= 2.612 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 57.8(1 \cdot 9 - 1 \cdot 8.29) - 2 \cdot 15.8$
 $(1 \cdot 9 - 1 \cdot 8.29) \cdot (1 - 0.83)$
 $= 0.375 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 22.5)(15.8 - 0.53)0.83/\sin(66.8)$
 $= 6.214 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$
 $= 0.833 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K7-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 \cdot Wo$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (2.61 - 0.38 + 2 \cdot 15.8 \cdot 0.83 \cdot$
 $(1 \cdot 9 - 8.29)) \cdot 138)$
 $= 3412.26 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can/0.707)^2 \cdot fr2) \cdot Sv$
 $= (6.21 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138$
 $= 9917.31 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (6.21 + 0 + 0.83 + (2.36)) \cdot 138$
 $= 13244.67 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (6.21 + 0 + 0.83 + 0 + (2.36)) \cdot 138$
 $= 13244.67 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 92.3 \cdot 10 \cdot 0.49 \cdot 115$
 $= 8332. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 37.5) \cdot (16.8 - 1) \cdot 0.7 \cdot 115$
 $= 15255. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 92.3 \cdot (10 - 1) \cdot 0.74 \cdot 138$
 $= 13589. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (8332 + 15255) = 23587 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (8332 + 0 + 13589 + 0) = 21921 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (8332 + 13589 + 0) = 21921 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 23587	Kgf, must exceed W = 3412 Kgf	or W1 = 9917 Kgf
Path 2-2 = 21921	Kgf, must exceed W = 3412 Kgf	or W2 = 13244 Kgf
Path 3-3 = 21921	Kgf, must exceed W = 3412 Kgf	or W3 = 13244 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.52 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K7-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.51 \cdot 4508 \cdot 0.9 / (2 \cdot 138 \cdot 1 \cdot 0.2 \cdot 3.51)$
 $= 5.1544 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R(\exp([P/(Sn \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4(\exp([3.51/(115 \cdot 1)] - 1)$
 $= 0.0806 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5345 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 105.6000 mm.
 Parallel to Vessel Wall, opening length d 52.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 22.5000 mm.

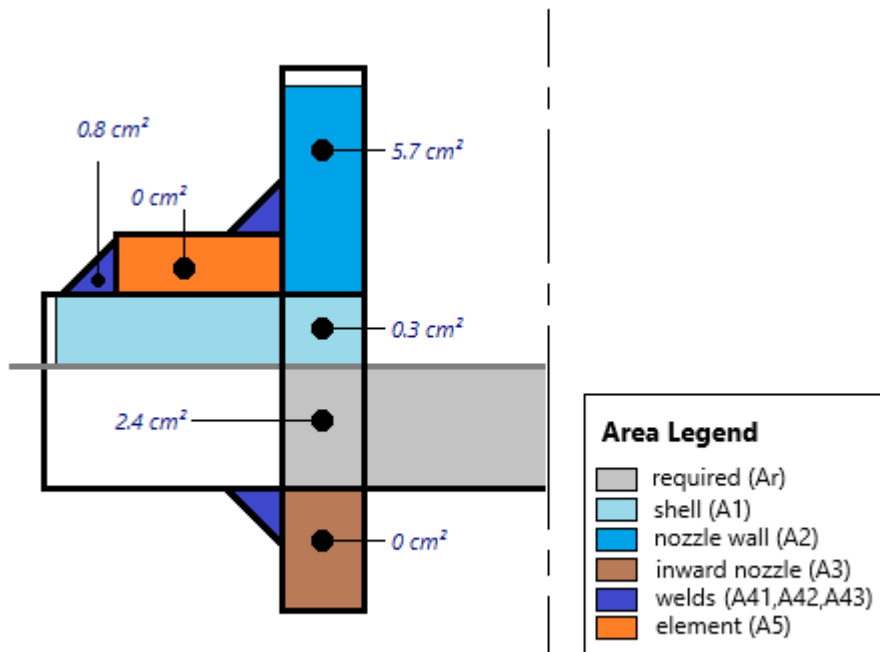


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	2.992	2.405	NA
Area in Shell	A1	1.828	0.339	NA
Area in Nozzle Wall	A2	5.881	5.711	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.833	0.833	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	8.543	6.884	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(52.8 \cdot 8.29 \cdot 1 + 2 \cdot 15.8 \cdot 8.29 \cdot 1(1 - 0.83))$
 $= 2.405 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$

$$= 52.8 (1 \cdot 9 - 1 \cdot 8.29) - 2 \cdot 15.8$$

$$(1 \cdot 9 - 1 \cdot 8.29) \cdot (1 - 0.83)$$

$$= 0.339 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_r^2$$

$$= (2 \cdot 22.5) (15.8 - 0.53) 0.83$$

$$= 5.711 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_r^2 + (W_i - \text{can}/0.707)^2 \cdot f_r^2$$

$$= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$$

$$= 0.833 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 1.5345 \text{ mm.}$
Wall Thickness per UG16(b),	$t_{r16b} = 2.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 6.7262 \text{ mm.}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 6.7262 \text{ mm.}$
Wall Thickness, shell/head, external pressure	$t_{rb2} = 1.8441 \text{ mm.}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 2.5000 \text{ mm.}$
Wall Thickness per table UG-45	$t_{b3} = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[5.8, \max(6.73, 2.5)]$$

$$= 5.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.53, 5.8)$$

$$= 5.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K7-2"

Intermediate Calc. for nozzle/shell welds $T_{min} = 11.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (2.41 - 0.34 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 9 - 8.29)) \cdot 138)$$

$$= 3171.04 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can}/0.707)^2 \cdot f_r^2) \cdot S_v$$

$$= (5.71 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138$$

$$= 9209.00 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (5.71 + 0 + 0.83 + (2.36)) \cdot 138$$

$$= 12536.37 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (5.71 + 0 + 0.83 + 0 + (2.36)) \cdot 138$$

$$= 12536.37 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115$$

$$= 7611. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115$$

$$= 13936. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 84.3 \cdot (10 - 1) \cdot 0.74 \cdot 138$$

$$= 12414. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (7611 + 13936) = 21547 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (7611 + 0 + 12414 + 0) = 20025 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (7611 + 12414 + 0) = 20025 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 21547	Kgf, must exceed W = 3171 Kg	or W1 = 9209 Kg
Path 2-2 = 20025	Kgf, must exceed W = 3171 Kg	or W2 = 12536 Kg
Path 3-3 = 20025	Kgf, must exceed W = 3171 Kg	or W3 = 12536 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 17.0214 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 777.7797 mm.

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Input, Nozzle Desc: K8-2"

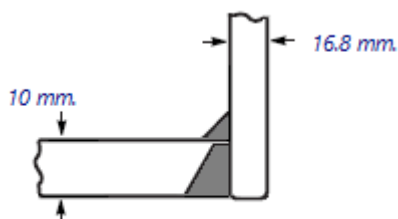
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Pressure for Reinforcement Calculations	P	3.507	bars
Temperature for Internal Pressure	Temp	110	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	4506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	750.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K8-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.51 \cdot 4508 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.51)$
 $= 5.1544 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 (\exp([3.51 / (115 \cdot 1)] - 1))$
 $= 0.0806 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5345 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	115.5970 mm.
Parallel to Vessel Wall, opening length	d	57.7985 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 115/138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 115/138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

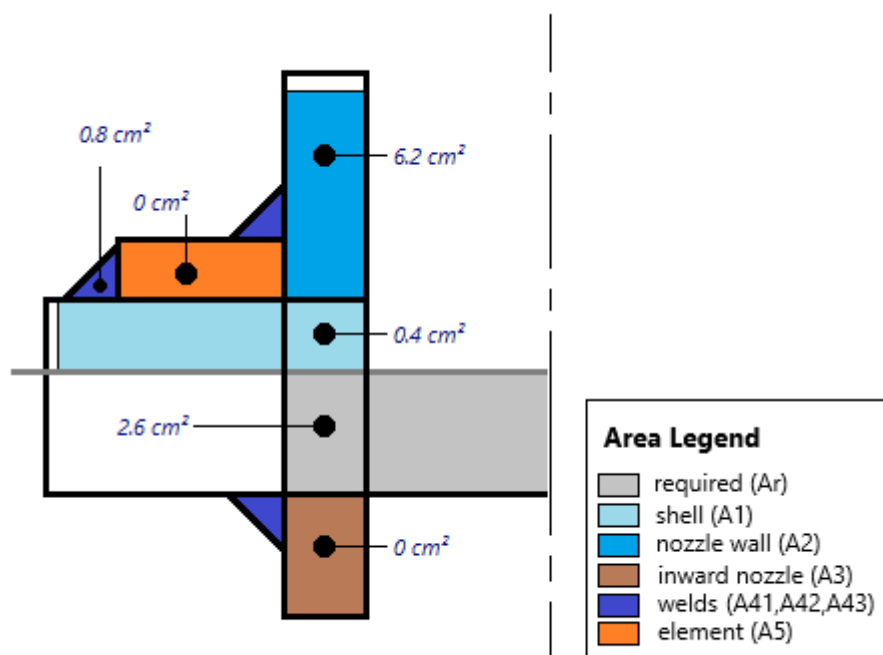


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3.250	2.612	NA
Area in Shell	A1	2.021	0.375	NA
Area in Nozzle Wall	A2	6.400	6.214	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.833	0.833	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	9.253	7.423	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 66.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(57.8 \cdot 8.29 \cdot 1 + 2 \cdot 15.8 \cdot 8.29 \cdot 1(1 - 0.83))$
 $= 2.612 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 57.8(1 \cdot 9 - 1 \cdot 8.29) - 2 \cdot 15.8$
 $(1 \cdot 9 - 1 \cdot 8.29) \cdot (1 - 0.83)$
 $= 0.375 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 22.5)(15.8 - 0.53)0.83/\sin(66.8)$
 $= 6.214 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$
 $= 0.833 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K8-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 \cdot Wo$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (2.61 - 0.38 + 2 \cdot 15.8 \cdot 0.83 \cdot$
 $(1 \cdot 9 - 8.29)) \cdot 138)$
 $= 3412.26 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can/0.707)^2 \cdot fr2) \cdot Sv$
 $= (6.21 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138$
 $= 9917.31 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (6.21 + 0 + 0.83 + (2.36)) \cdot 138$
 $= 13244.67 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (6.21 + 0 + 0.83 + 0 + (2.36)) \cdot 138$
 $= 13244.67 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 92.3 \cdot 10 \cdot 0.49 \cdot 115$
 $= 8332. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 37.5) \cdot (16.8 - 1) \cdot 0.7 \cdot 115$
 $= 15255. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 92.3 \cdot (10 - 1) \cdot 0.74 \cdot 138$
 $= 13589. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (8332 + 15255) = 23587 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (8332 + 0 + 13589 + 0) = 21921 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (8332 + 13589 + 0) = 21921 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 23587	Kgf, must exceed W = 3412 Kgf	or W1 = 9917 Kgf
Path 2-2 = 21921	Kgf, must exceed W = 3412 Kgf	or W2 = 13244 Kgf
Path 3-3 = 21921	Kgf, must exceed W = 3412 Kgf	or W3 = 13244 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.52 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K8-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.51 \cdot 4508 \cdot 0.9 / (2 \cdot 138 \cdot 1 \cdot 0.2 \cdot 3.51)$
 $= 5.1544 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R(\exp([P/(Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4(\exp([3.51/(115 \cdot 1)] - 1))$
 $= 0.0806 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5345 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 105.6000 mm.
 Parallel to Vessel Wall, opening length d 52.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 22.5000 mm.

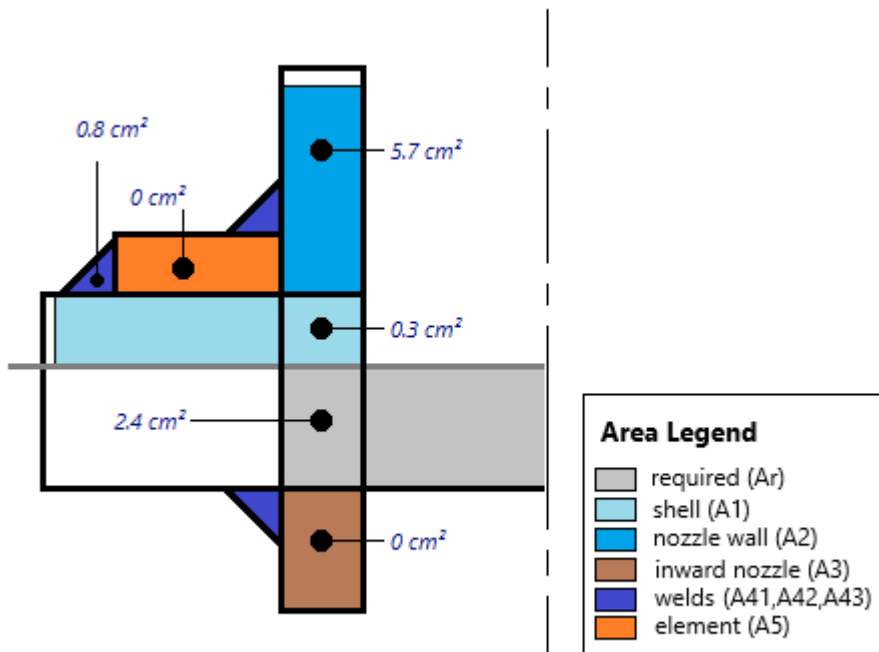


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	2.992	2.405	NA
Area in Shell	A1	1.828	0.339	NA
Area in Nozzle Wall	A2	5.881	5.711	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.833	0.833	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	8.543	6.884	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(52.8 \cdot 8.29 \cdot 1 + 2 \cdot 15.8 \cdot 8.29 \cdot 1(1 - 0.83))$
 $= 2.405 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$

$$= 52.8 (1 \cdot 9 - 1 \cdot 8.29) - 2 \cdot 15.8$$

$$(1 \cdot 9 - 1 \cdot 8.29) \cdot (1 - 0.83)$$

$$= 0.339 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f r_2$$

$$= (2 \cdot 22.5) (15.8 - 0.53) 0.83$$

$$= 5.711 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f r_2 + (W_i - \text{can}/0.707)^2 \cdot f r_2$$

$$= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$$

$$= 0.833 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 1.5345 \text{ mm.}$
Wall Thickness per UG16(b),	$t_{r16b} = 2.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 6.7262 \text{ mm.}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 6.7262 \text{ mm.}$
Wall Thickness, shell/head, external pressure	$t_{rb2} = 1.8441 \text{ mm.}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 2.5000 \text{ mm.}$
Wall Thickness per table UG-45	$t_{b3} = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[5.8, \max(6.73, 2.5)]$$

$$= 5.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.53, 5.8)$$

$$= 5.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K8-2"

Intermediate Calc. for nozzle/shell welds	T_{min}	11.0000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) S_v)$$

$$= \max(0, (2.41 - 0.34 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 9 - 8.29)) 138)$$

$$= 3171.04 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can}/0.707)^2 \cdot f r_2) \cdot S_v$$

$$= (5.71 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138$$

$$= 9209.00 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v$$

$$= (5.71 + 0 + 0.83 + (2.36)) \cdot 138$$

$$= 12536.37 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (5.71 + 0 + 0.83 + 0 + (2.36)) \cdot 138$$

$$= 12536.37 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= $(\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
= $(3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115$
= 7611. Kgf

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115$
= 13936. Kgf

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
= $(3.14/2.0) \cdot 84.3 \cdot (10 - 1) \cdot 0.74 \cdot 138$
= 12414. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (7611 + 13936) = 21547 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (7611 + 0 + 12414 + 0) = 20025 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (7611 + 12414 + 0) = 20025 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 21547	Kgf, must exceed W = 3171 Kgf	or W1 = 9209 Kgf
Path 2-2 = 20025	Kgf, must exceed W = 3171 Kgf	or W2 = 12536 Kgf
Path 3-3 = 20025	Kgf, must exceed W = 3171 Kgf	or W3 = 12536 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.52 bars

The Drop for this Nozzle is : 17.0214 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 777.7797 mm.

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: